MIL-M-28787/296A(EC)
14 July 1980
SUPERSEDING
MIL-M-28787/296(EC)
4 December 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, J$\bar{K}$, DIGITAL, KEY CODE GHK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic J$\bar{K}$ flip-flop module.

* 1.2 Classification. This specification covers two classes of modules.

Class I and class II as specificed in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.094 failures per million operating hours for class I or 0.282 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GHK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/296-1 for class I and M28787/296-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent.

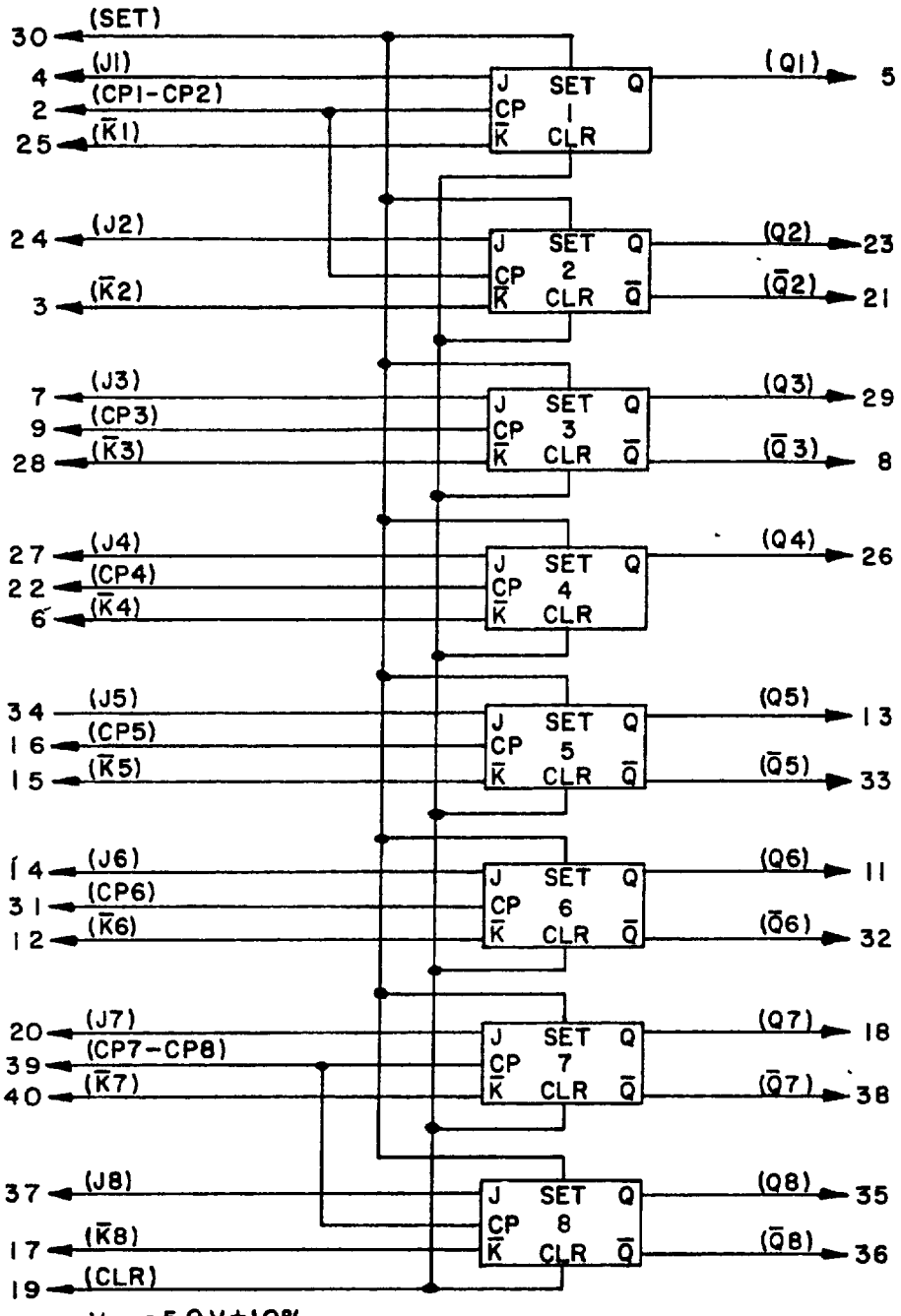
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTES:
1. J1-J8, K̄1-K̄8 = Data inputs.
2. CP1-CP2, CP3-CP6, CP7-CP8 = Clock inputs.
3. SET = Common set input.
4. CLR = Common clear input.
5. Q1-Q8 = Data outputs.
6. Q̄2, Q̄3, Q̄5-Q̄8 = Data outputs.

FIGURE 1. GHK module function diagram, eight JK̄ FLIP-FLOPS.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial End-of-life	III	S	 10 1	 ∞ ∞	 10 1	 ∞ ∞	 10 1	 ∞ ∞	 10 1	 ∞ ∞	 10 1	 ∞ ∞	 MΩ MΩ
B1-B14	V_{OUT} (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1,C2, C25,C26	Input breakdown current	V	S	0.0	200	0.0	200	0.0	200	0.0	200	0.0	200	µA
C3-C22, C27-C30	Input breakdown current	V	S	0.0	100	0.0	100	0.0	100	0.0	100	0.0	100	µA
C23,C24	Input breakdown current	V	S	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	mA
D1-D14	Output short circuit current	VI	X	-6	-100	-6	-100	-6	-100	-6	-100	-6	-100	mA
E1-E14	Output leakage current	VII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G8	Functional test	IX	X	← Output must be as specified in table IX →										
H1-H14	Static output voltage, low level	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J14	Static output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1, K4	Input current, high level	XI	X	0	80	0	80	0	80	0	80	0	80	µA
K2,K3, K25,K26	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	µA
K5-K24, K27-K30	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
L1,L4	Input current, low level	XII	X	-0.1	-1.72	-0.1	-1.72	-0.1	-1.72	-0.1	-1.72	-0.1	-1.72	mA
L2,L3	Input current, low level	XII	X	-0.1	-0.87	-0.1	-0.87	-0.1	-0.87	-0.1	-0.87	-0.1	-0.87	mA
L5-L24	Input current, low level	XII	X	-0.1	-0.43	-0.1	-0.43	-0.1	-0.43	-0.1	-0.43	-0.1	-0.43	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M	Power supply current	XIII	S	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	mA
N1-N14	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	10	50	10	50	10	55	10	50	10	60	ns
N15-N28	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	20	80	20	80	20	90	20	80	20	100	ns
P1-P14	Transition time (t_{THL}), C_L = 50 pF	XIV	S	3	18	3	18	3	20	3	18	3	22	ns
P15-P28	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	6	40	6	40	6	42	6	40	6	44	ns
Q1-Q14	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	XV	S	3	29	3	29	3	31	3	29	3	33	ns
Q15-Q28	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	XV	Q	4	39	4	39	4	41	4	39	4	43	ns
R1-R14	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	XV	S	4	44	4	44	4	46	4	44	4	48	ns
R15-R28	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	XV	Q	5	54	5	54	5	56	5	54	5	58	ns
S1-S8	Propagation delay time from set to Q (t_{PLH}), C_L = 50 pF	XVI	S	5	55	5	55	5	57	5	55	5	59	ns
S9-S16	Propagation delay time from set to Q (t_{PLH}), C_L = 150 pF	XVI	Q	5	65	5	65	5	67	5	65	5	69	ns
T1-T6	Propagation delay time from set to $\overline{Q}$ (t_{PHL}), C_L = 50 pF	XVI	S	6	65	6	65	6	67	6	65	6	71	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T7-T12	Propagation delay time from set to $\overline{Q}$ (t_{PHL}), C_L = 150 pF	XVI	Q	7	75	7	75	7	77	7	75	7	81	ns
U1-U6	Propagation delay time from clear to $\overline{Q}$ (t_{PLH}), C_L = 50 pF	XVI	S	5	50	5	50	5	52	5	50	5	56	ns
U7-U12	Propagation delay time from clear to $\overline{Q}$ (t_{PLH}), C_L = 150 pF	XVI	Q	6	60	6	60	6	62	6	60	6	66	ns
V1-V8	Propagation delay time from clear to Q (t_{PHL}) C_L = 50 pF	XVI	S	6	63	6	63	6	67	6	63	6	71	ns
V9-V16	Propagation delay time from clear to Q (t_{PHL}) C_L = 150 pF	XVI	Q	7	73	7	73	7	77	7	73	7	81	ns
W1-W14	Maximum operating frequency check, C_L = 50 pF	XVII	S	75	85	75	85	75	85	75	85	75	85	ns
X1-X14	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y1-Y14	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
BB1	Input capacitance	XXI	Q	5	45	5	45	5	45	5	45	5	45	pF
BB2-BB4	Input capacitance	XXI	Q	5	30	5	30	5	30	5	30	5	30	pF
BB5-BB24	Input capacitance	XXI	Q	5	15	5	15	5	15	5	15	5	15	pF
CC1-CC28	Setup time	XXII	S	170	230	170	230	170	230	170	230	170	230	ns
DD1-DD28	Hold time	XXIII	S	170	230	170	230	170	230	170	230	170	230	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. If test conditions A,B,C, or D are part of a sequence of tests, the order of testing shall be as follows: A,B,C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through DD in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA, CC and DD shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25°±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through AA, CC and DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module function. This module contains eight $J\overline{K}$ FLIP-FLOPS with common buffered asynchronous SET and CLEAR inputs (active high) for all FLIP-FLOPS. The clock input is common to FLIP-FLOPS 1 and 2 (CP1-CP2) and to FLIP-FLOPS 7 and 8 (CP7-CP8). FLIP-FLOPS 3-6 each have an independent clock input. FLIP-FLOPS 2, 3, 5, 6, 7, and 8 have both a Q and $\overline{Q}$ output. FLIP-FLOPS 1 and 4 have only a Q output. Information on the inputs is transferred to the output on the positive clock input transition.

* 6.4.1 Input to output relationship. Each $J\overline{K}$ FLIP-FLOP has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Inputs					Outputs	
SET	CLEAR	CLOCK	J	$\overline{K}$	Q	$\overline{Q}$
H	L	X	X	X	H	L
L	H	X	X	X	L	H
H	H	X	X	X	H	H
		↑			(see note 6)	(see note 6)
L	L	↑	L	L	L	H
L	L	↑	H	L	Toggle	
L	L	↑	L	H	Q_o	$\overline{Q}_o$
L	L	↑	H	H	H	L
L	L	L	X	X	Q_o	$\overline{Q}_o$

NOTES:
1. H = High-level voltage (steady state).
2. L = Low-level voltage (steady state).
3. X = Irrelevant (any input, including transitions).
4. ↑ = Transition from low to high.
5. Q_o, $\overline{Q}_o$ = The level of Q, $\overline{Q}$ respectively, before the indicated steady-state input conditions were established.
6. This configuration is unstable; that is, it will not persist when SET and CLEAR return to their inactive (low-level) state.

* 6.4.2 Parallel outputs. Paralleling of outputs is not permissible.

* 6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life parameters from table I.

	Class I	Class II	Unit
Power dissipation	212 max	212 max	mW
Input voltage			
Low level	-0.5 min, 0.7 max	-0.5 min, 0.7 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current, CLR			
Low level (V_{IL}=0.0 V)	-1.72 max	-1.72 max	mA
High level (V_{IH}=2.7 V)	80 max	80 max	µA
(V_{IH}=5.5V)	1.6 max	1.6 max	mA
Input current, CP1,2,7,8			
Low level (V_{IL}=0.0 V)	-0.87 max	-0.87 max	mA
High level (V_{IH}=2.7 V)	40 max	40 max	µA
(V_{IH}=5.5 V)	0.2 max	0.2 max	mA
Input current, CP3-6			
Low level (V_{IL}=0.0 V)	-0.43 max	-0.43 max	mA
High level (V_{IH}=2.7 V)	20 max	20 max	µA
(V_{IH}=5.5 V)	0.1 max	0.1 max	mA

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	Class I	Class II	Unit
Input current, SET			
Low level (V_{IL}=0.0 V)	-1.72 max	-1.72 max	mA
High level (V_{IH}=2.7 V)	80 max	80 max	µA
(V_{IH}=5.5 V)	1.6 max	1.6 max	mA
Input current, $J\overline{K}$			
Low level (V_{IL}=0.0 V)	-0.43 max	-0.43 max	mA
High level (V_{IH}=2.7 V)	20 max	20 max	µA
(V_{IH}=5.5 V)	0.1 max	0.1 max	mA
Output drive current			
Low level (V_{OL}=0.4 V)	4.0 max	4.0 max	mA
High level(V_{OH}=2.5 V)	-400 mA	-400 mA	µA
Operating frequency	25 max	25 max	MHz
Setup time			
$t_{setup}(J, \overline{K})$	20 min	20 min	ns
Hold time			
$t_{hold}(J, \overline{K})$	5 min	5 min	ns
Pulse width			
t_{PW} (CLOCK)	25 min	25 min	ns
t_{PW} (SET, CLR)	25 min	25 min	ns

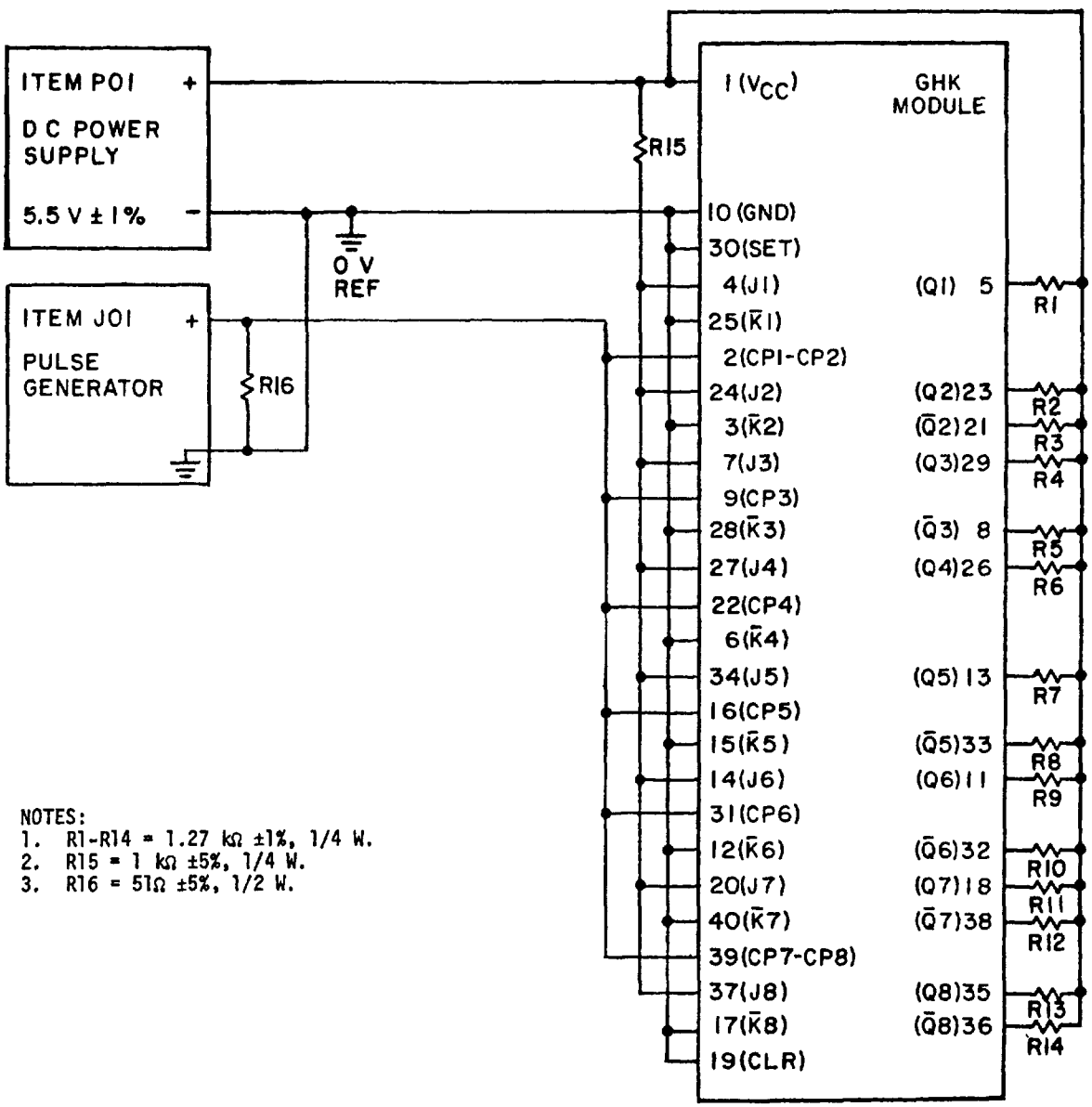
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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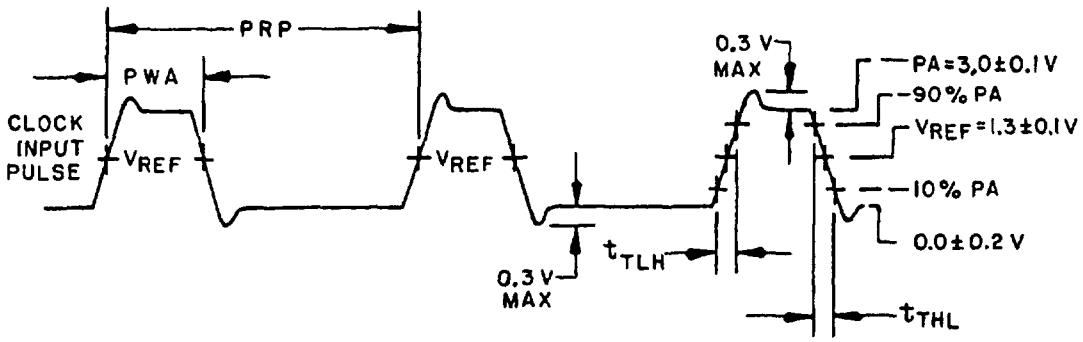
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NOTES:
1. R1-R14 = 1.27 kΩ ±1%, 1/4 W.
2. R15 = 1 kΩ ±5%, 1/4 W.
3. R16 = 51Ω ±5%, 1/2 W.

FIGURE 2. Operating temperature and life test circuit.

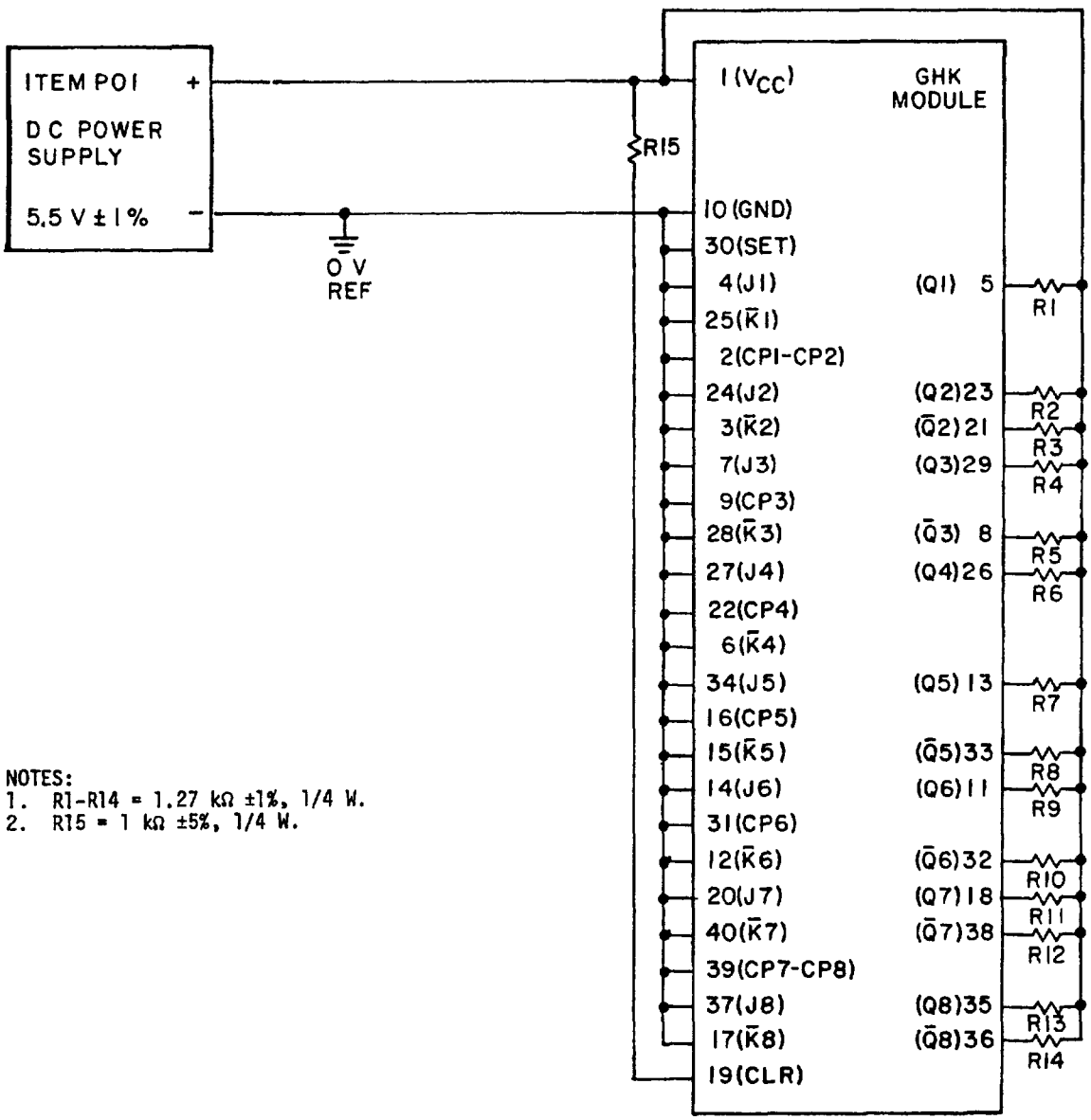
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltages are referenced to ground.

FIGURE 3. Operating temperature and life test circuit waveforms.

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NOTES:
1. R1-R14 = 1.27 kΩ ±1%, 1/4 W.
2. R15 = 1 kΩ ±5%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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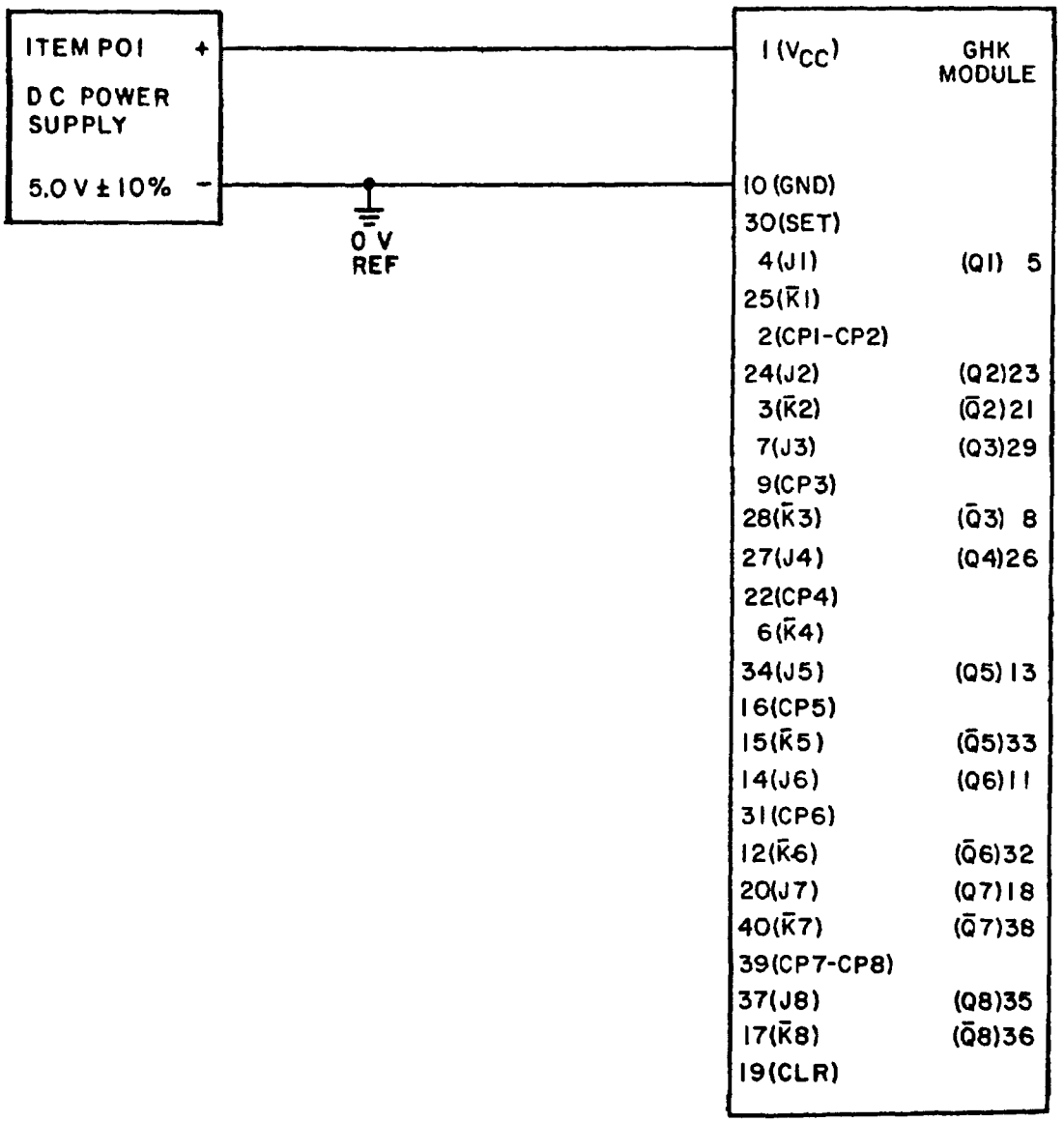


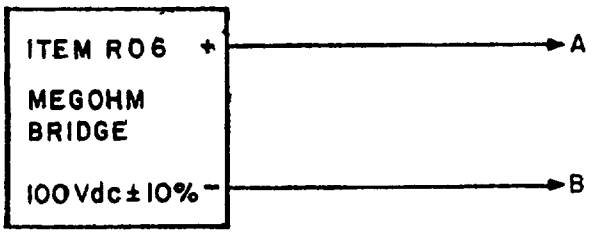
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	6

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I (V_{CC})	GHK MODULE
10 (GND)	
30(SET)	
4(J1)	(Q1) 5
25($\bar{K}$1)	
2(CP1-CP2)	
24(J2)	(Q2)23
3($\bar{K}$2)	($\bar{Q}$2)21
7(J3)	(Q3)29
9(CP3)	
28($\bar{K}$3)	($\bar{Q}$3) 8
27(J4)	(Q4)26
22(CP4)	
6($\bar{K}$4)	
34(J5)	(Q5)13
16(CP5)	
15($\bar{K}$5)	($\bar{Q}$5)33
14(J6)	(Q6)11
31(CP6)	
12($\bar{K}$6)	($\bar{Q}$6)32
20(J7)	(Q7)18
40($\bar{K}$7)	($\bar{Q}$7)38
39(CP7-CP8)	
37(J8)	(Q8)35
17($\bar{K}$8)	($\bar{Q}$8)36
19(CLR)	

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

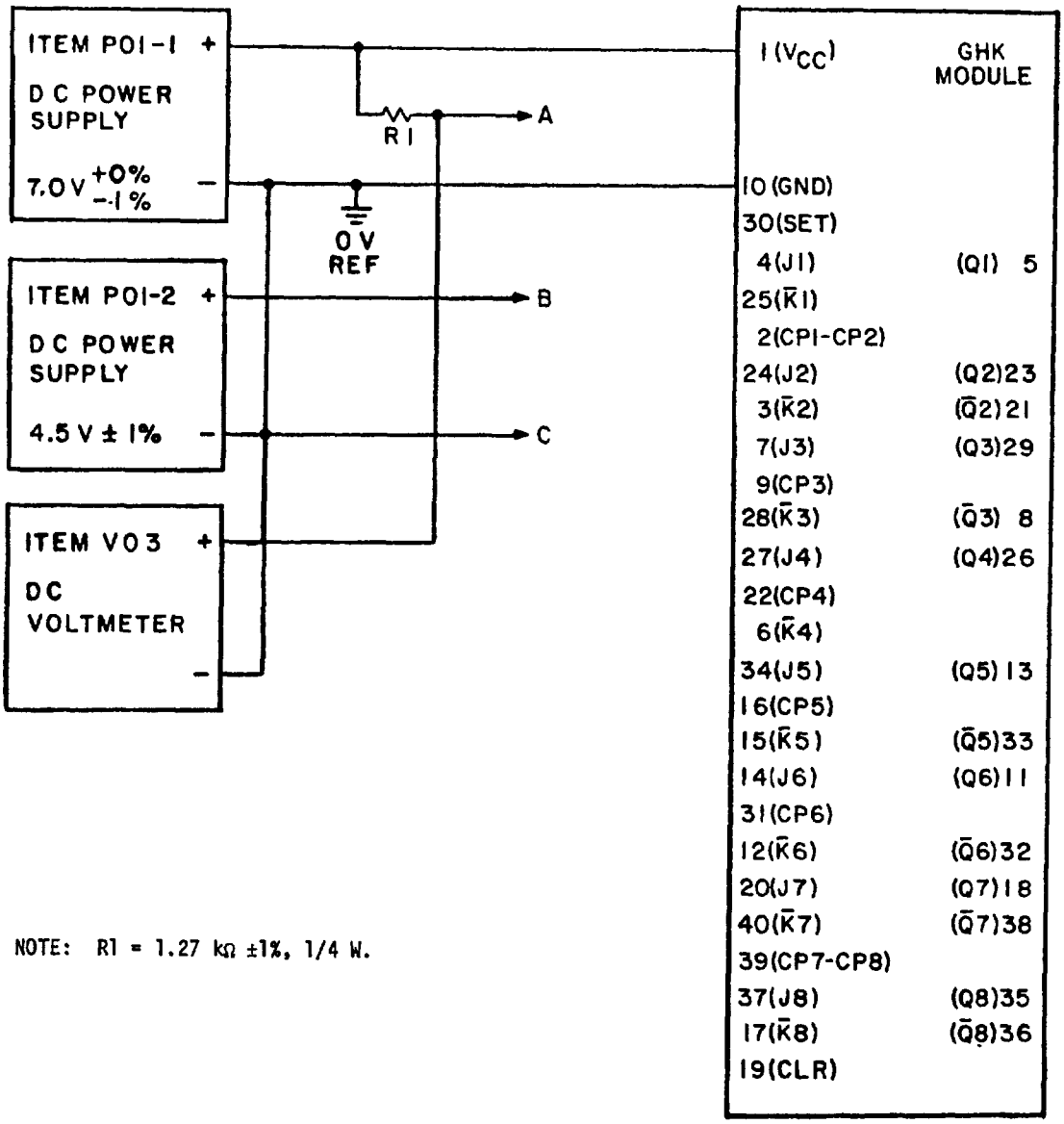
FIGURE 6. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	Q1 V_{OUT} (low-loaded)	5	19	30	0.01	0.8	V	7
B2	Q2	23						
B3	Q3	29						
B4	Q4	26						
B5	Q5	13						
B6	Q6	11						
B7	Q7	18						
B8	Q8	35	19	30				
B9	$\overline{Q}$2	21	30	19				
B10	$\overline{Q}$3	8						
B11	$\overline{Q}$5	33						
B12	$\overline{Q}$6	32						
B13	$\overline{Q}$7	38						
B14	$\overline{Q}$8 V_{OUT} (low-loaded)	36	30	19	0.01	0.8	V	7

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NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	CP1-CP2	Input breakdown current	3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	2	0.0	200	µA	8
C2	CP7-CP8		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,40	39		200		
C3	CP3		2,3,4,6,7,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	9		100		
C4	CP4		2,3,4,6,7,9,12,14,15,16, 17,20,24,25,27,28,30,31, 34,37,39,40	22				
C5	CP5		2,3,4,6,7,9,12,14,15,17, 20,22,24,25,27,28,30,31, 34,37,39,40	16				
C6	CP6		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 34,37,39,40	31				
C7	J1		2,3,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	4				
C8	J2		2,3,4,6,7,9,12,14,15,16, 17,20,22,25,27,28,30,31, 34,37,39,40	24				
C9	J3	Input breakdown current	2,3,4,6,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	7	0.0	100	µA	8

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C10	J4	Input breakdown current	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,28,30,31, 34,37,39,40	27	0.0	100	µA	8
C11	J5		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,37,39,40	34				
C12	J6		2,3,4,6,7,9,12,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	14				
C13	J7		2,3,4,6,7,9,12,14,15,16, 17,22,24,25,27,28,30,31, 34,37,39,40	20				
C14	J8		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,39,40	37				
C15	$\overline{K1}$		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,27,28,30,31, 34,37,39,40	25				
C16	$\overline{K2}$		2,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	3				
C17	$\overline{K3}$		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,30,31, 34,37,39,40	28				
C18	$\overline{K4}$	Input breakdown current	2,3,4,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	6	0.0	100	µA	8

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TABLE V. Input breakdown current test - Continued.

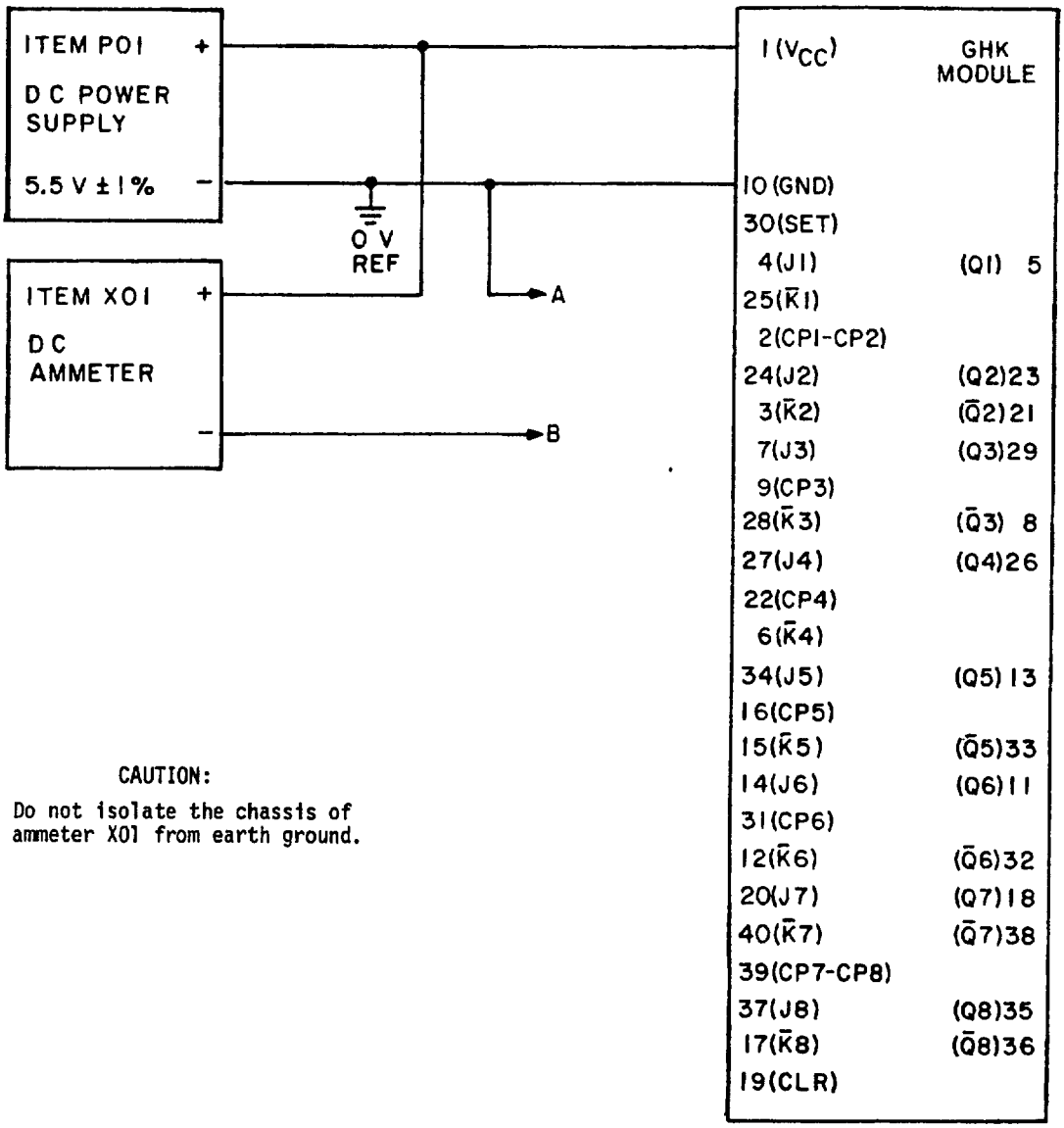
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C19	$\overline{K5}$	Input breakdown current	2,3,4,6,7,9,12,14,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	15	0.0	100	µA	8
C20	$\overline{K6}$		2,3,4,6,7,9,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	12				
C21	$\overline{K7}$		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,39	40				
C22	$\overline{K8}$		2,3,4,6,7,9,12,14,15,16, 20,22,24,25,27,28,30,31, 34,37,39,40	17		100	µA	
C23	CLR		2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,39,40	19		1.6	mA	
C24	SET		2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 31,34,37,39,40	30		1.6	mA	
C25	CP1-CP2		3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	2		200	µA	
C26	CP7-CP8		2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 31,34,37,40	39		200	µA	
C27	CP3	Input breakdown current	2,3,4,6,7,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	9	0.0	100	µA	8

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C28	CP4 Input breakdown current	2,3,4,6,7,9,12,14,15,16, 17,19,20,24,25,27,28,31, 34,37,39,40	22	0.0	100	µA	8
C29	CP5 Input breakdown current	2,3,4,6,7,9,12,14,15,17, 19,20,22,24,25,27,28,31, 34,37,39,40	16	0.0	100	µA	8
C30	CP6 Input breakdown current	2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 34,37,39,40	31	0.0	100	µA	8

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

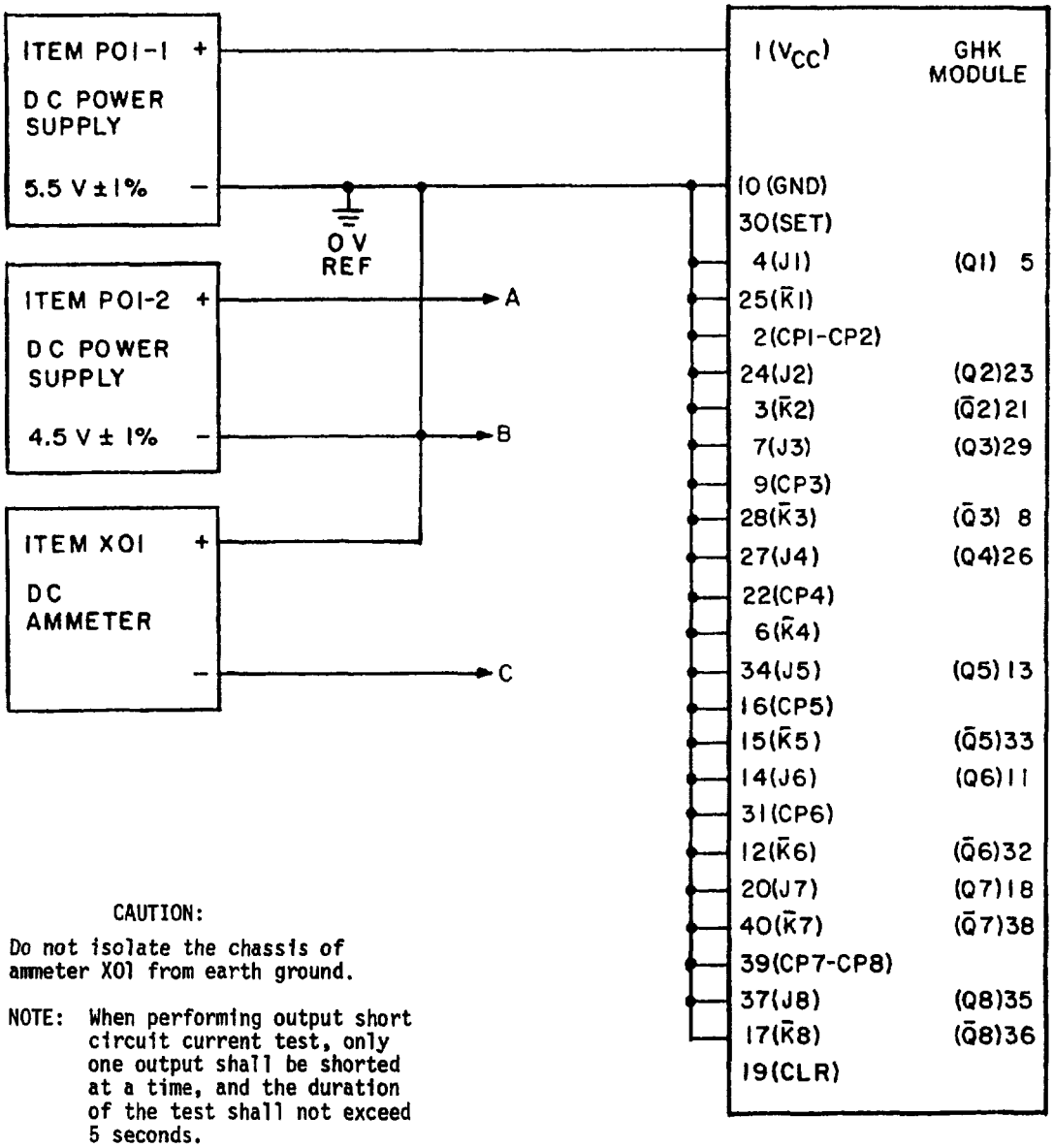
FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Q1 Output short circuit current	30	19	5	-6	-100	mA	9
D2	Q2			23				
D3	Q3			29				
D4	Q4			26				
D5	Q5			13				
D6	Q6			11				
D7	Q7			18				
D8	Q8	30	19	35				
D9	$\overline{Q2}$	19	30	21				
D10	$\overline{Q3}$			8				
D11	$\overline{Q5}$			33				
D12	$\overline{Q6}$			32				
D13	$\overline{Q7}$			38				
D14	$\overline{Q8}$ Output short circuit current	19	30	36	-6	-100	mA	9

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

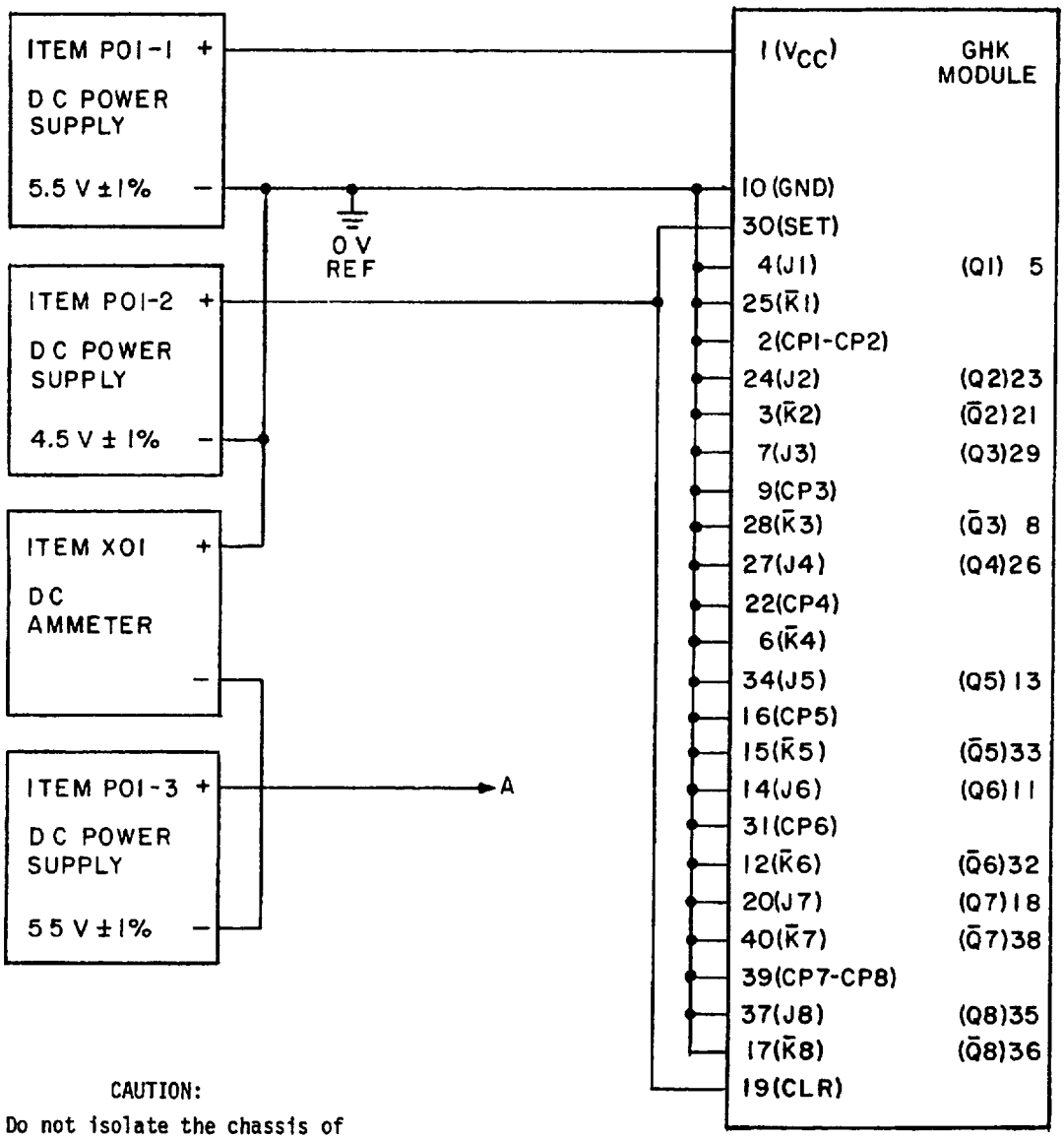
FIGURE 9. Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
E1	Q1	Output leakage current	5	0.0	1.0	mA	10
E2	Q2	↓	23	↓	↓	↓	↓
E3	$\overline{Q2}$		21				
E4	Q3		29				
E5	$\overline{Q3}$		8				
E6	Q4		26				
E7	Q5		13				
E8	$\overline{Q5}$		33				
E9	Q6		11				
E10	$\overline{Q6}$		32				
E11	Q7		18				
E12	$\overline{Q7}$		38				
E13	Q8		35				
E14	$\overline{Q8}$	Output leakage current	36	0.0	1.0	mA	10

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P01-3 shall be current limited to 5.0 mA.

FIGURE 10. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

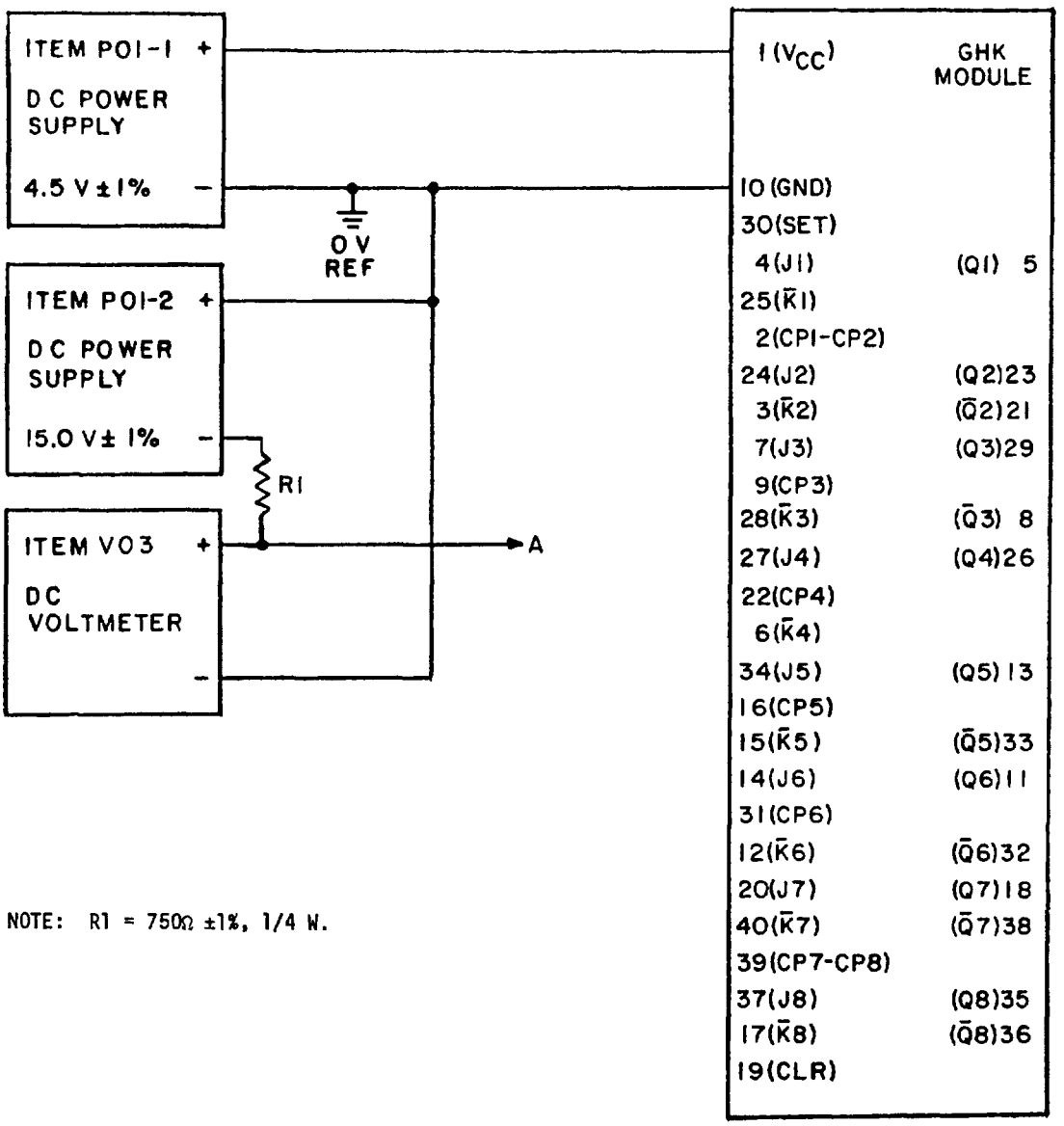
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
F1	SET	Input clamp diode voltage	30	-0.4	-1.5	V	11
F2	CLR		19				
F3	CP1-CP2		2				
F4	CP3		9				
F5	CP4		22				
F6	CP5		16				
F7	CP6		31				
F8	CP7-CP8		39				
F9	J1		4				
F10	J2		24				
F11	J3		7				
F12	J4		27				
F13	J5		34				
F14	J6		14				
F15	J7		20				
F16	J8		37				
F17	$\overline{K1}$		25				
F18	$\overline{K2}$	Input clamp diode voltage	3	-0.4	-1.5	V	11

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TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F19	$\overline{K}3$ Input clamp diode voltage	28	-0.4	-1.5	V	11
F20	$\overline{K}4$	6	↓	↓	↓	↓
F21	$\overline{K}5$	15	↓	↓	↓	↓
F22	$\overline{K}6$	12	↓	↓	↓	↓
F23	$\overline{K}7$	40	↓	↓	↓	↓
F24	$\overline{K}8$ Input clamp diode voltage	17	-0.4	-1.5	V	11

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NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VIII.

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* TABLE IX. Functional test.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q5}$	$\overline{Q6}$	$\overline{Q7}$	$\overline{Q8}$	Min	Max		
G1	Functional	19,4,25,24, 3,7,28,27,6, 34,15,14,12, 20,40,37,17	30	See notes 2 and 3	2,9,22,16, 31,39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	See note 1			12,13
G2		30	19,4,25,24, 3,7,28,27, 6,34,15,14, 12,20,40, 37,17			L	L	L	L	L	L	L	L	H	H	H	H	H	H				
G3		Open	30,19,4,25, 24,7,28,27, 6,34,15,14, 12,20,40, 37,17,3			H	H	H	H	H	H	H	H	H	H	H	H	H	H				
G4	Functional	30,19	4,25,24,3, 7,28,27,6, 34,15,14, 12,20,40, 37,17	See notes 2 and 3	2,9,22,16, 31,39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	See note 1			12,13

See notes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q5}$	$\overline{Q6}$	$\overline{Q7}$	$\overline{Q8}$	Min	Max		
G5	Functional	30,19,4,25, 24,3,7,28, 27,6,34,15, 14,12,20,40, 37,17	Open	See notes 2 and 3	2,9,22,16, 31,39	L	L	L	L	L	L	L	L	H	H	H	H	H	H	See note 1			12,13
G6		30,19,4,24, 7,27,34,14, 20,37	25,3,28,6, 15,12,40, 17			L	L	L	L	L	L	L	L	H	H	H	H	H	H				
G7		30,19,25,3, 28,6,15,12, 40,17	4,24,7,27, 34,14,20, 37			H	H	H	H	H	H	H	H	L	L	L	L	L	L				
G8	Functional	30,19,4,24, 7,27,34,14, 20,37	25,3,28,6, 15,12,40, 17	See notes 2 and 3	2,9,22,16, 31,39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	See note 1			12,13

NOTES:

1. Output logic states:

   a. Low level 0.0 to 0.4 V.
   b. High level 2.5 to 4.5 V.

2. For test conditions G1-G8, verification of the output states in note 1 is obtained by connecting voltmeter V03, point C to each output pin in succession after input conditions have been accomplished. Test conditions G1-G8 are to be performed in sequence.
3. Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, $\overline{Q2}$, $\overline{Q3}$, $\overline{Q5}$, $\overline{Q6}$, $\overline{Q7}$, $\overline{Q8}$ output logic states are measured at output pins 5, 23, 29, 26, 13, 11, 18, 35, 21, 8, 33, 32, 38, 36.

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TABLE X. Static output voltage (low-level and high-level) test and input threshold check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Q1 Static output voltage, low level	30	19	5	Open	0.0	0.4	V	12
H2	Q2			23					
H3	Q3			29					
H4	Q4			26					
H5	Q5			13					
H6	Q6			11					
H7	Q7			18					
H8	Q8	30	19	35					
H9	$\overline{Q2}$	19	30	21					
H10	$\overline{Q3}$			8					
H11	$\overline{Q5}$			33					
H12	$\overline{Q6}$			32					
H13	$\overline{Q7}$			38					
H14	$\overline{Q8}$ Static output voltage, low level	19	30	36	Open	0.0	0.4	V	12

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TABLE X. Static output voltage (low-level and high-level) test and input threshold check - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Q1 Static output voltage, high level	3,6,7, 12,14, 15,17, 19,20, 24,27, 28,30, 34,37, 40	4,25	5	2,9,16, 22,31, 39	2.5	4.5	V	12,13
J2	Q2	4,6,7, 12,14, 15,17, 19,20, 25,27, 28,30, 34,37, 40	3,24	23					
J3	Q3	3,4,6, 12,14, 15,17, 19,20, 24,25, 27,30, 34,37, 40	7,28	29					
J4	Q4 Static output voltage, high level	3,4,7, 12,14, 15,17, 19,20, 24,25, 28,30, 34,37, 40	6,27	26	2,9,16, 22,31, 39	2.5	4.5	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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TABLE X. Static output voltage (low-level and high-level) test and input threshold check - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J5	Q5 Static output voltage, high level	3,4,6, 7,12, 14,17, 19,20, 24,25, 27,28, 30,37, 40	15,34	13	2,9,16, 22,31, 39	2.5	4.5	V	12,13
J6	Q6	3,4,6, 7,15, 17,19, 20,24, 25,27, 28,30, 34,37, 40	12,14	11					
J7	Q7	3,4,6, 7,12, 14,15, 17,19, 24,25, 27,28, 30,34, 37	20,40	18					
J8	Q8 Static output voltage, high level	3,4,6, 7,12, 14,15, 19,20, 24,25, 27,28, 30,34, 40	17,37	35	2,9,16, 22,31, 39	2.5	4.5	V	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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TABLE X. Static output voltage (low-level and high-level) test and input threshold check - Continued.

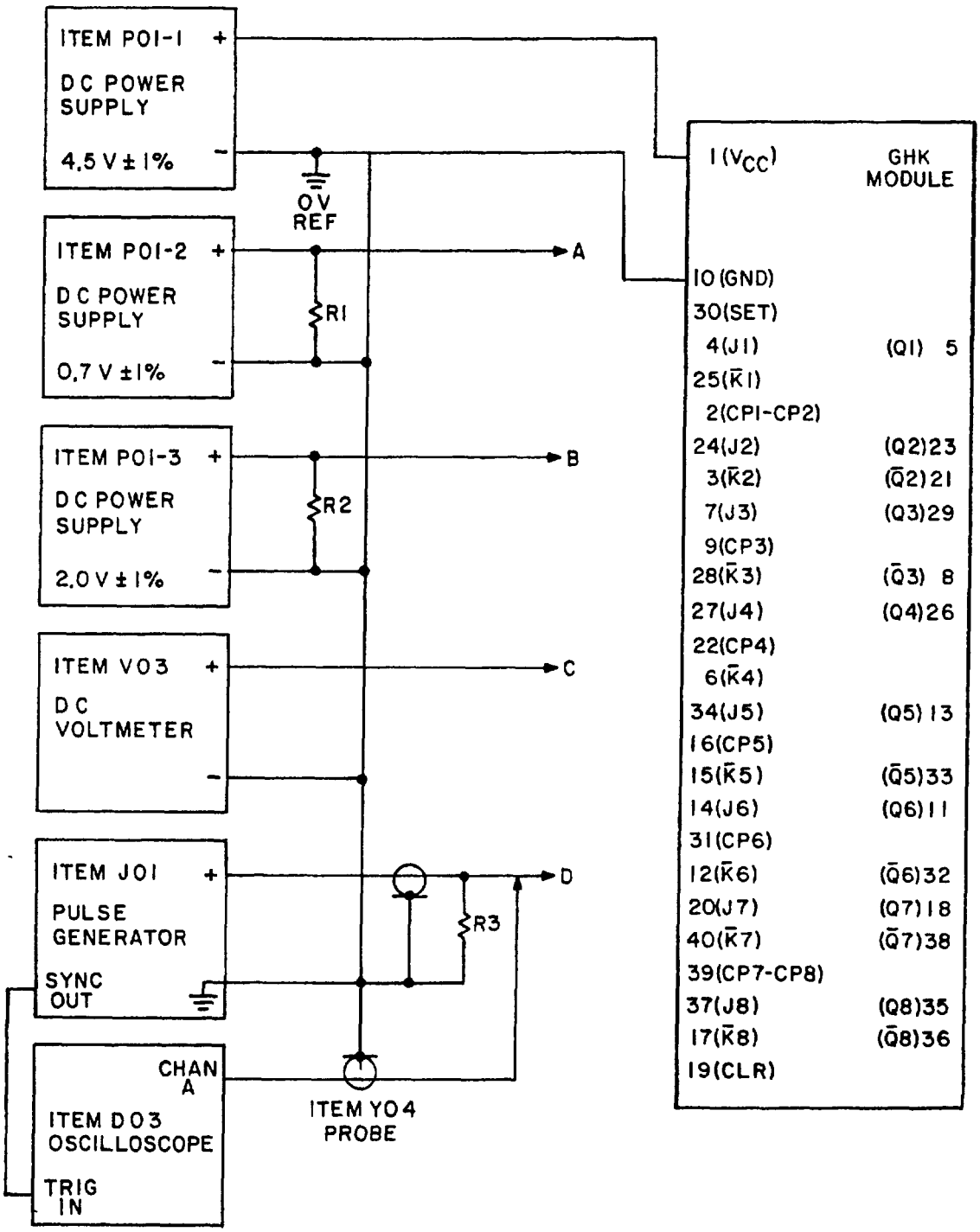
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J9	$\overline{Q}2$ Static output voltage, high level	3,19, 24,30	4,6,7, 12,14, 15,17, 20,25, 27,28, 34,37, 40	21	2,9,16, 22,31, 39	2.5	4.5	V	12,13
J10	$\overline{Q}3$	7,19, 28,30	3,4,6, 12,14, 15,17, 20,24, 25,27, 34,37, 40	8					
J11	$\overline{Q}5$	15,19, 30,34	3,4,6, 7,12, 14,17, 20,24, 25,27, 28,37, 40	33					
J12	$\overline{Q}6$ Static output voltage, high level	12,14, 19,30	3,4,6, 7,15, 17,20, 24,25, 27,28, 34,37, 40	32	2,9,16, 22,31, 39	2.5	4.5	V	12,13

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TABLE X. Static output voltage (low-level and high-level) test and input threshold check - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J13	$\overline{Q7}$ Static output voltage, high level	19,20, 30,40	3,4,6, 7,12, 14,15, 17,24, 25,27, 28,34, 37	38	2,9,16, 22,31, 39	2.5	4.5	V	12,13
J14	$\overline{Q8}$ Static output voltage, high level	17,19, 30,37	3,4,6, 7,12, 14,15, 20,24, 25,27, 28,34, 40	36	2,9,16, 22,31, 39	2.5	4.5	V	12,13

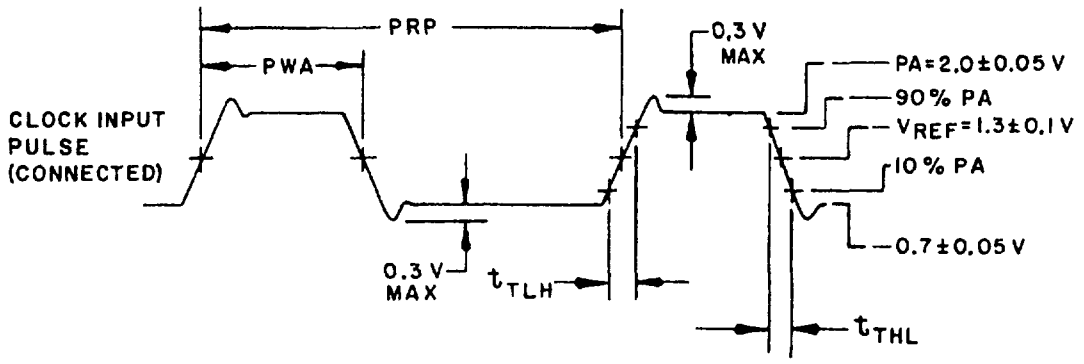
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NOTES:
1. R1, R2, R3 = 51Ω ±5%, 1/2 W.
2. For test conditions G1-G8 and J1-J14, make all input circuit connections for a test condition then use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
3. For test conditions H1-H14 and J1-J14, load the output under test and the corresponding complement output (where available) with the load circuit per figure 14. Load circuits are not required for test conditions G1-G8.

FIGURE 12. Functional, static output voltage (low-level and high-level), and input threshold check test circuit for tables IX and X.

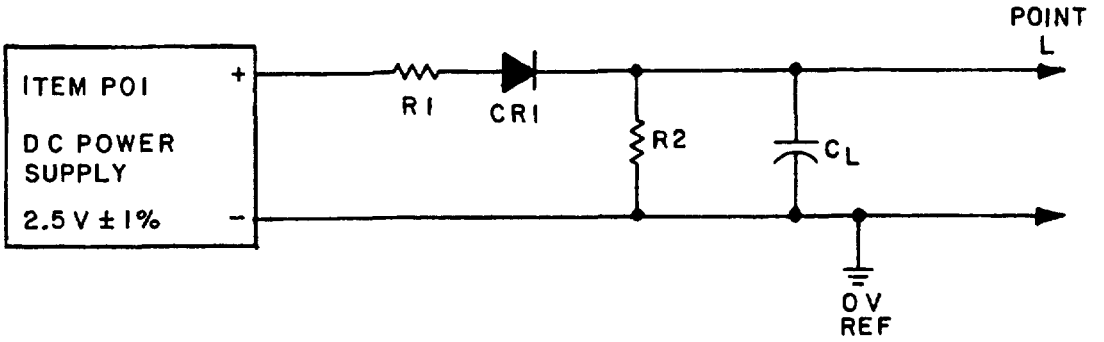
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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
2. Clock threshold level setup:
   With the clock input connected, place item J01 in the single pulse mode. Use item D03 to adjust dc offset of item J01 for a low level of 0.7 ±0.05 V (clock low level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D03 to adjust item J01 amplitude for a high level of 2.0 ±0.05 V (clock high level threshold).

FIGURE 13. Functional and static output voltage (high-level) test and input threshold check test circuit waveform for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 4.0 mA ±5% with point L held at 0.4 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. Test type S:

   C_L = 50 ±5 pF (including probe and parasitic capacitance).

   Test type Q:

   C_L = 150 ±5 pF (including probe and parasitic capacitance).

5. Capacitor C_L is not required for dc electrical test conditions H1-H14 and J1-J14.
6. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 14. GHK logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	CLR Input current, high level	19	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	0	80	µA	15
K2	CP1-CP2	2	3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40		40		
K3	CP7-CP8	39	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,40		40		
K4	SET	30	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40		80		
K5	CP3	9	2,3,4,6,7,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40		20		
K6	CP4	22	2,3,4,6,7,9,12,14,15,16,17, 20,24,25,27,28,30,31,34,37, 39,40				
K7	CP5	16	2,3,4,6,7,9,12,14,15,17,20, 22,24,25,27,28,30,31,34,37, 39,40				
K8	CP6	31	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,34,37, 39,40				
K9	J1 Input current, high level	4	2,3,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	0	20	µA	15

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K10	J2	Input current, high level	24	2,3,4,6,7,9,12,14,15,16,17, 20,22,25,27,28,30,31,34,37, 39,40	0	20	µA	15
K11	J3		7	2,3,4,6,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40				
K12	J4		27	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,28,30,31,34,37, 39,40				
K13	J5		34	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,37, 39,40				
K14	J6		14	2,3,4,6,7,9,12,15,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40				
K15	J7		20	2,3,4,6,7,9,12,14,15,16,17, 22,24,25,27,28,30,31,34,37, 39,40				
K16	J8		37	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 39,40				
K17	$\overline{K1}$		25	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,27,28,30,31,34,37, 39,40				
K18	$\overline{K2}$	Input current, high level	3	2,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	0	20	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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TABLE XI. Input current (high-level) test - Continued.

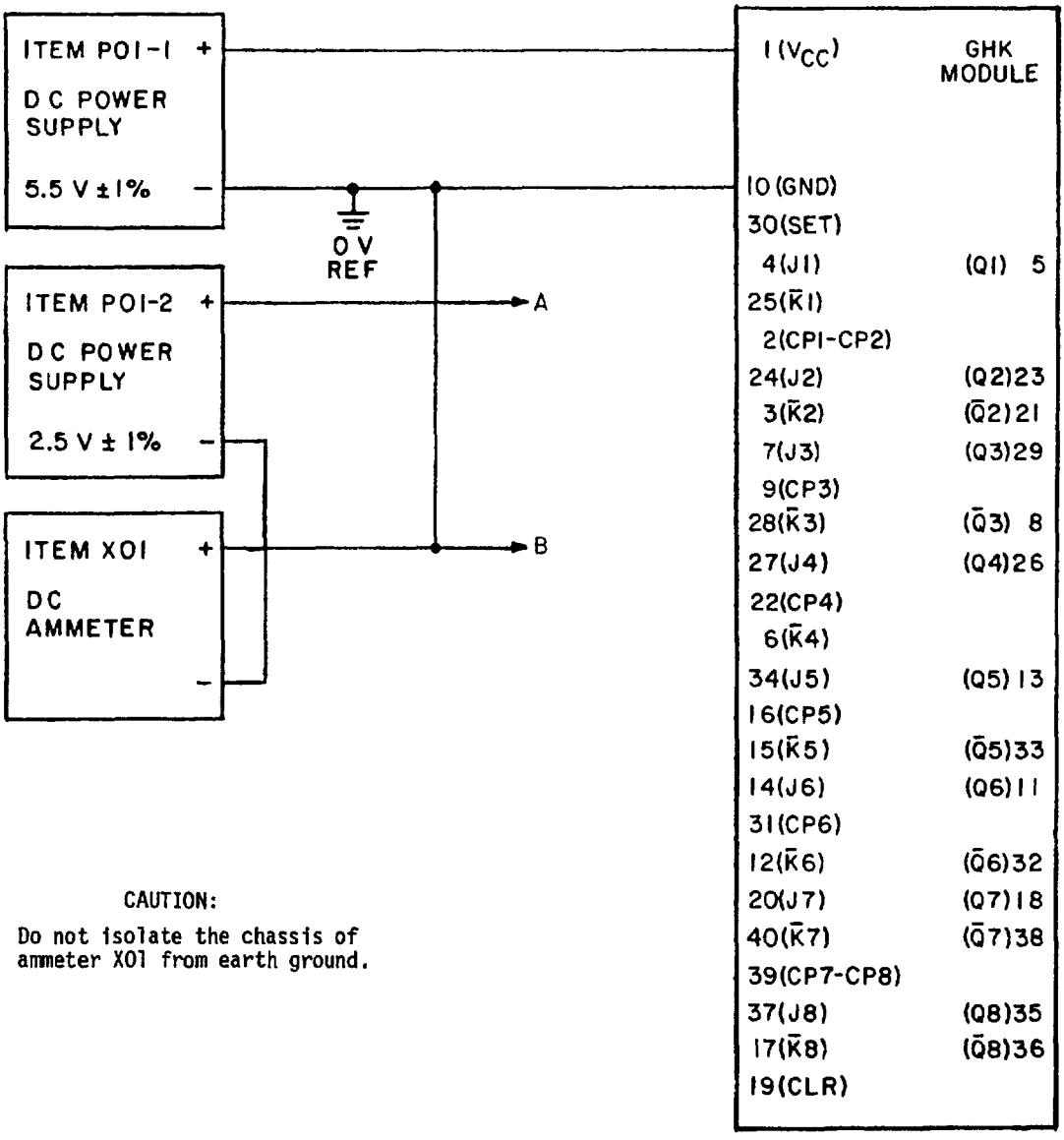
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K19	$\overline{K3}$ Input current, high level	28	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,30,31,34,37, 39,40	0	20	µA	15
K20	$\overline{K4}$	6	2,3,4,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40				
K21	$\overline{K5}$	15	2,3,4,6,7,9,12,14,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40				
K22	$\overline{K6}$	12	2,3,4,6,7,9,14,15,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40				
K23	$\overline{K7}$	40	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39				
K24	$\overline{K8}$	17	2,3,4,6,7,9,12,14,15,16,20, 22,24,25,27,28,30,31,34,37, 39,40		20		
K25	CP1-CP2	2	3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40		40		
K26	CP7-CP8	39	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,40		40		
K27	CP3 Input current, high level	9	2,3,4,6,7,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40	0	20	µA	15

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K28	CP4 Input current, high level	22	2,3,4,6,7,9,12,14,15,16,17, 19,20,24,25,27,28,31,34,37, 39, 40	0	20	µA	15
K29	CP5 Input current, high level	16	2,3,4,6,7,9,12,14,15,17,19, 20,22,24,25,27,28,31,34,37, 39,40	0	20	µA	15
K30	CP6 Input current, high level	31	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,34,37, 39,40	0	20	µA	15

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 15. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

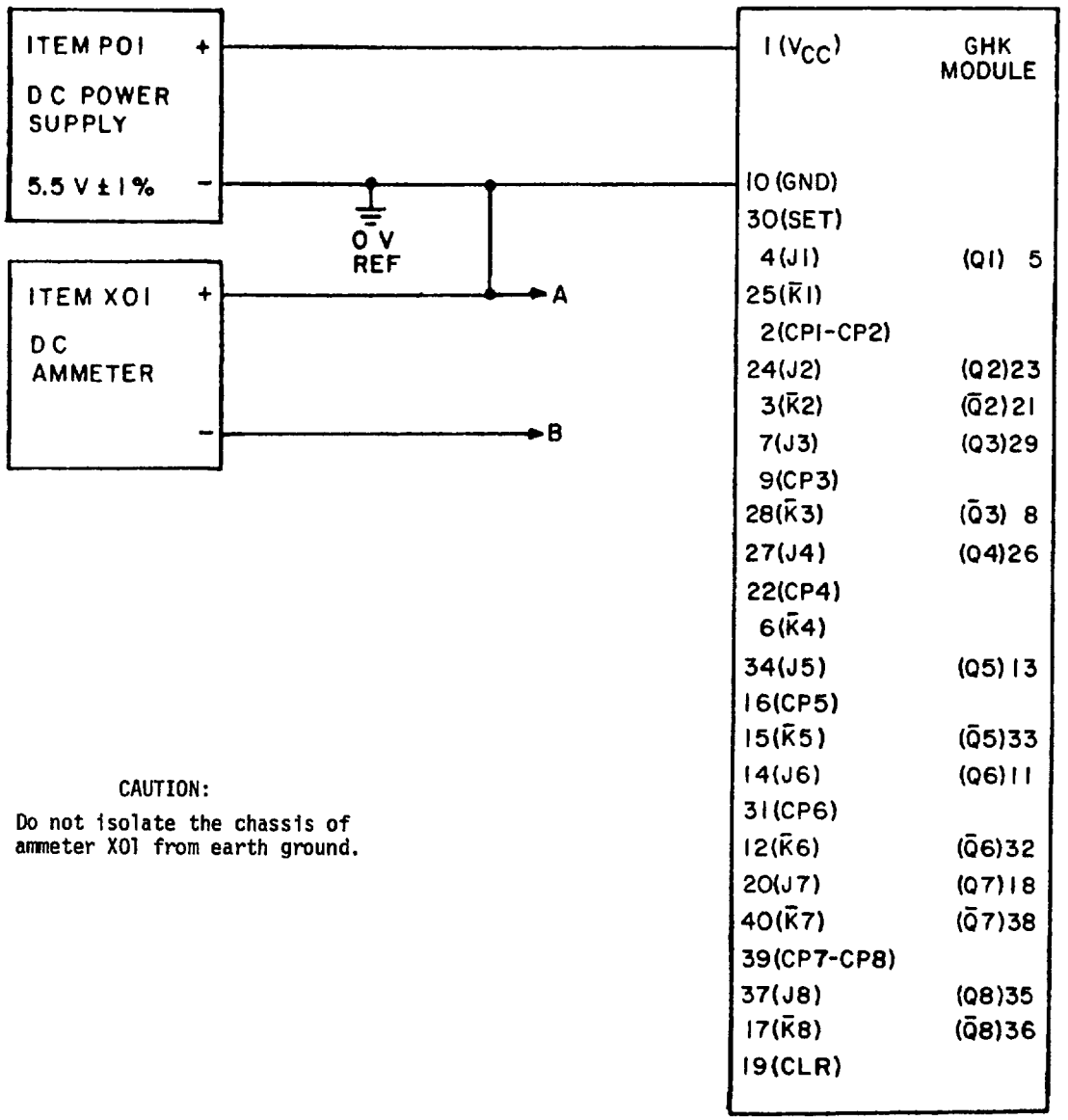
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	CLR Input current, low level	Open	19	-0.1	-1.72	mA	16
L2	CP1-CP2	3,4,19,24,25	2		-0.87		
L3	CP7-CP8	17,19,20,37, 40	39		-0.87		
L4	SET	Open	30		-1.72		
L5	CP3	7,19,28	9		-0.43		
L6	CP4	6,19,27	22				
L7	CP5	15,19,34	16				
L8	CP6	12,14,19	31				
L9	J1	19,30	4				
L10	J2		24				
L11	J3		7				
L12	J4		27				
L13	J5		34				
L14	J6		14				
L15	J7		20				
L16	J8	19,30	37				
L17	$\overline{K1}$	19	25				
L18	$\overline{K2}$ Input current, low level	19	3	-0.1	-0.43	mA	16

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L19	$\overline{K3}$ Input current, low level	19	28	-0.1	-0.43	mA	16
L20	$\overline{K4}$		6				
L21	$\overline{K5}$		15				
L22	$\overline{K6}$		12				
L23	$\overline{K7}$		40				
L24	$\overline{K8}$ Input current, low level	19	17	-0.1	-0.43	mA	16

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 16. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current	0.1	38.6	mA	17

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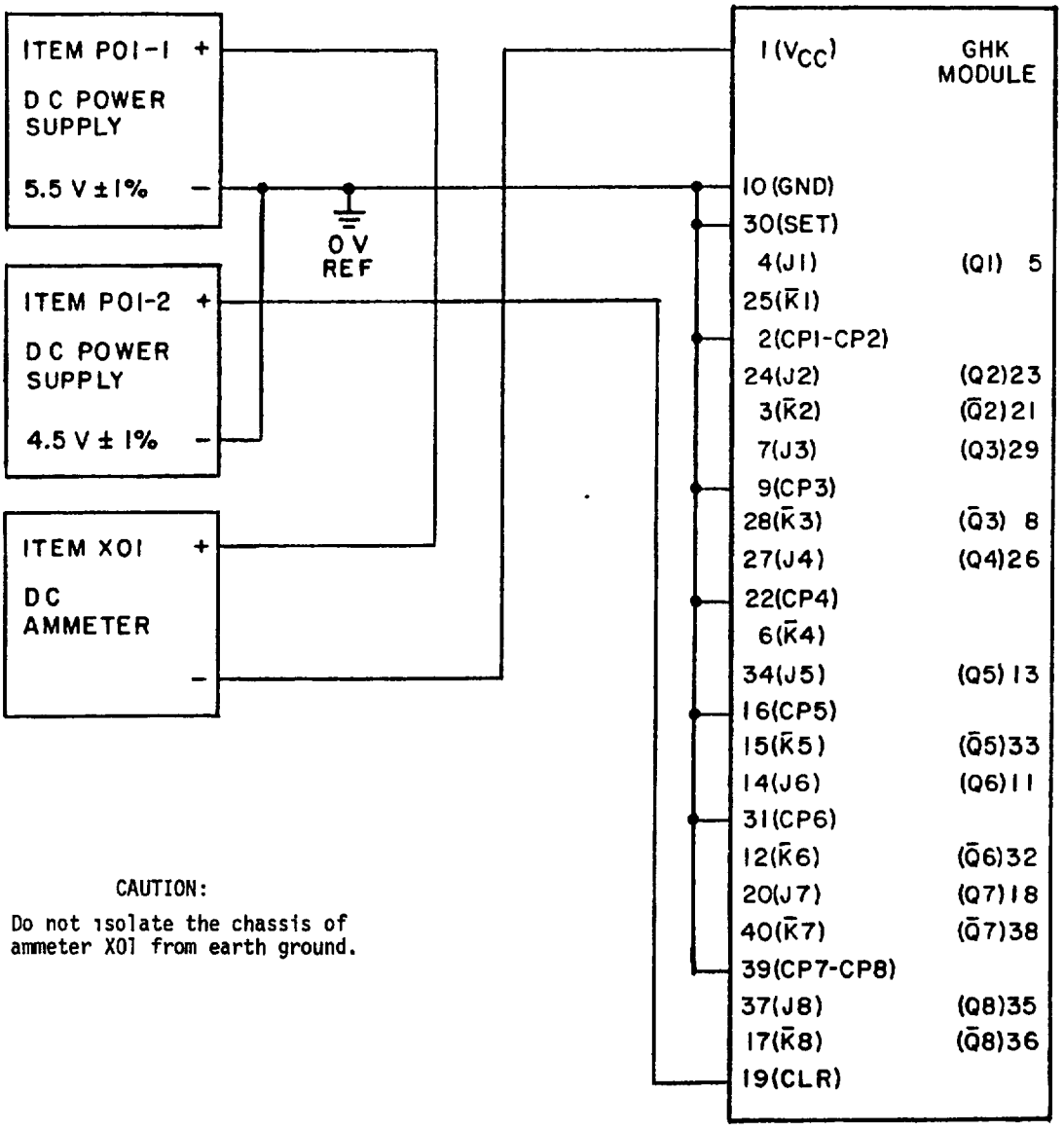


FIGURE 17. Power supply current test circuit for table XIII.

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* TABLE XIV. Transition time test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N1	Q1	Transition time (t_{TLH}), C_L = 50 pF	4,25	2	5	10	50	ns	18,19
N2	Q2		3,24	2	23				
N3	Q3		7,28	9	29				
N4	Q4		6,27	22	26				
N5	Q5		15,34	16	13				
N6	Q6		12,14	31	11				
N7	Q7		20,40	39	18				
N8	Q8		17,37	39	35				
N9	$\overline{Q2}$		3,24	2	21				
N10	$\overline{Q3}$		7,28	9	8				
N11	$\overline{Q5}$		15,34	16	33				
N12	$\overline{Q6}$		12,14	31	32				
N13	$\overline{Q7}$		20,40	39	38				
N14	$\overline{Q8}$	Transition time (t_{TLH}), C_L = 50 pF	17,37	39	36	10	50		
N15	Q1	Transition time (t_{TLH}), C_L = 150 pF	4,25	2	5	20	80		
N16	Q2	Transition time (t_{TLH}), C_L = 150 pF	3,24	2	23	20	80	ns	18,19

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* TABLE XIV. Transition time test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N17	Q3	Transition time (t_{TLH}), C_L = 150 pF	7,28	9	29	20	80	ns	18,19
N18	Q4	↓	6,27	22	26	↓	↓	↓	↓
N19	Q5	↓	15,34	16	13	↓	↓	↓	↓
N20	Q6	↓	12,14	31	11	↓	↓	↓	↓
N21	Q7	↓	20,40	39	18	↓	↓	↓	↓
N22	Q8	↓	17,37	39	35	↓	↓	↓	↓
N23	$\overline{Q2}$	↓	3,24	2	21	↓	↓	↓	↓
N24	$\overline{Q3}$	↓	7,28	9	8	↓	↓	↓	↓
N25	$\overline{Q5}$	↓	15,34	16	33	↓	↓	↓	↓
N26	$\overline{Q6}$	↓	12,14	31	32	↓	↓	↓	↓
N27	$\overline{Q7}$	↓	20,40	39	38	↓	↓	↓	↓
N28	$\overline{Q8}$	Transition time (t_{TLH}), C_L = 150 pF	17,37	39	36	20	80	↓	↓
P1	Q1	Transition time (t_{THL}), C_L = 50 pF	4,25	2	5	3	18	↓	↓
P2	Q2	↓	3,24	2	23	↓	↓	↓	↓
P3	Q3	↓	7,28	9	29	↓	↓	↓	↓
P4	Q4	Transition time (t_{THL}), C_L = 50 pF	6,27	22	26	3	18	ns	18,19

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* TABLE XIV. Transition time test - Continued.

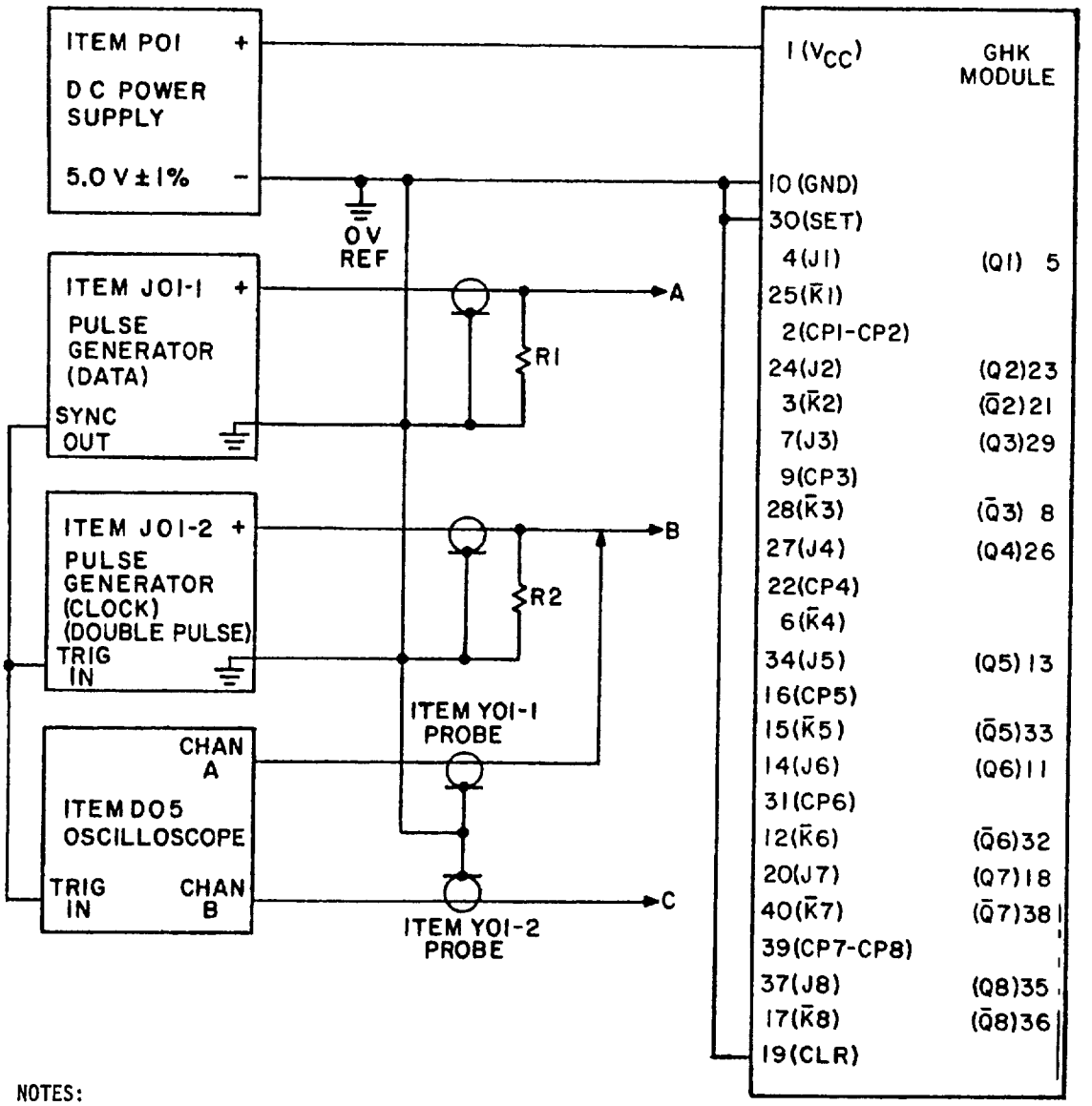
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P5	Q5	Transition time (t_{THL}), C_L = 50 pF	15,34	16	13	3	18	ns	18,19
P6	Q6		12,14	31	11				
P7	Q7		20,40	39	18				
P8	Q8		17,37	39	35				
P9	$\overline{Q2}$		3,24	2	21				
P10	$\overline{Q3}$		7,28	9	8				
P11	$\overline{Q5}$		15,34	16	33				
P12	$\overline{Q6}$		12,14	31	32				
P13	$\overline{Q7}$		20,40	39	38				
P14	$\overline{Q8}$	Transition time (t_{THL}), C_L = 50 pF	17,37	39	36	3	18		
P15	Q1	Transition time (t_{THL}), C_L = 150 pF	4,25	2	5	6	40		
P16	Q2		3,24	2	23				
P17	Q3		7,28	9	29				
P18	Q4		6,27	22	26				
P19	Q5		15,34	16	13				
P20	Q6	Transition time (t_{THL}), C_L = 150 pF	12,14	31	11	6	40	ns	18,19

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* TABLE XIV. Transition time test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P21	Q7	Transition time (t_{THL}), C_L = 150 pF	20,40	39	18	6	40	ns	18,19
P22	Q8	↓	17,37	39	35	↓	↓	↓	↓
P23	$\overline{Q2}$		3,24	2	21				
P24	$\overline{Q3}$		7,28	9	8				
P25	$\overline{Q5}$		15,34	16	33				
P26	$\overline{Q6}$		12,14	31	32				
P27	$\overline{Q7}$		20,40	39	38				
P28	$\overline{Q8}$	Transition time (t_{THL}), C_L = 150 pF	17,37	39	36	6	40	ns	18,19

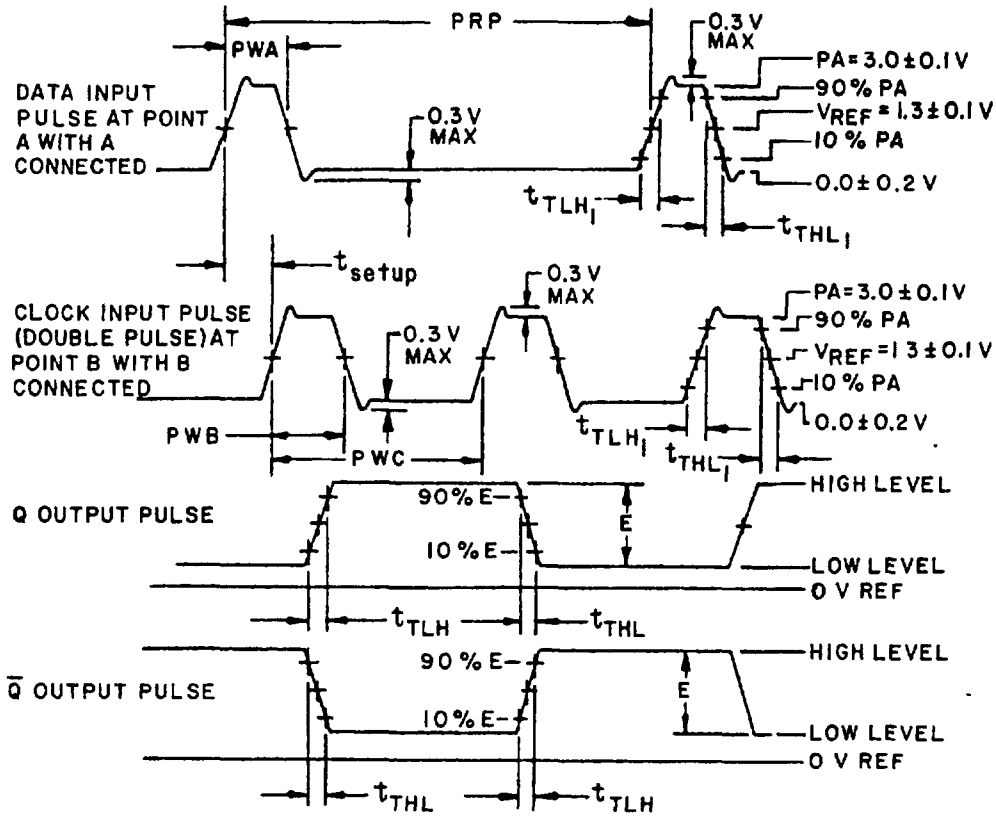
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 14.

FIGURE 18. Transition time test circuit for table XIV.

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NOTES:
1. Data input pulse parameters:

   a. PRP = 1,000 ±100 ns.
   b. $t_{TLHI} = t_{THLI}$ = 6 ±1 ns.
   c. PWA = 500 ±50 ns.

2. Clock input pulse parameters:

   a Item J01-2 is operated in double pulse mode.
   b. PWB = 200 ±20 ns.
   c. $t_{TLHI} = t_{THLI}$ = 6 ±1 ns.
   d. Use delay control of item J01-2 to obtain PWC = 500 ±50 ns.
   e. t_{setup} = 100 ±10 ns.

3. Verify that all outputs have a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements per table XIV.
4. All voltages are referenced to ground.

FIGURE 19. Transition time test circuit waveforms for table XIV.

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* TABLE XV. Propagation delay time from clock test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
Q1	Q1	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	4	25	2	5	3	29	ns	20,21
Q2	Q2		24	3	2	23				
Q3	Q3		7	28	9	29				
Q4	Q4		27	6	22	26				
Q5	Q5		34	15	16	13				
Q6	Q6		14	12	31	11				
Q7	Q7		20	40	39	18				
Q8	Q8		37	17	39	35				
Q9	$\overline{Q2}$		24	3	2	21				
Q10	$\overline{Q3}$		7	28	9	8				
Q11	$\overline{Q5}$		34	15	16	33				
Q12	$\overline{Q6}$		14	12	31	32				
Q13	$\overline{Q7}$		20	40	39	38				
Q14	$\overline{Q8}$	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	37	17	39	36	3	29		
Q15	Q1	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	4	25	2	5	4	39		
Q16	Q2	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	24	3	2	23	4	39	ns	20,21

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* TABLE XV. Propagation delay time from clock test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
Q17	Q3	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	7	28	9	29	4	39	ns	20,21
Q18	Q4		27	6	22	26				
Q19	Q5		34	15	16	13				
Q20	Q6		14	12	31	11				
Q21	Q7		20	40	39	18				
Q22	Q8		37	17	39	35				
Q23	$\overline{Q2}$		24	3	2	21				
Q24	$\overline{Q3}$		7	28	9	8				
Q25	$\overline{Q5}$		34	15	16	33				
Q26	$\overline{Q6}$		14	12	31	32				
Q27	$\overline{Q7}$		20	40	39	38				
Q28	$\overline{Q8}$	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	37	17	39	36		39		
R1	Q1	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	4	25	2	5		44		
R2	Q2		24	3	2	23				
R3	Q3		7	28	9	29				
R4	Q4	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	27	6	22	26	4	44	ns	20,21

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* TABLE XV. Propagation delay time from clock test - Continued.

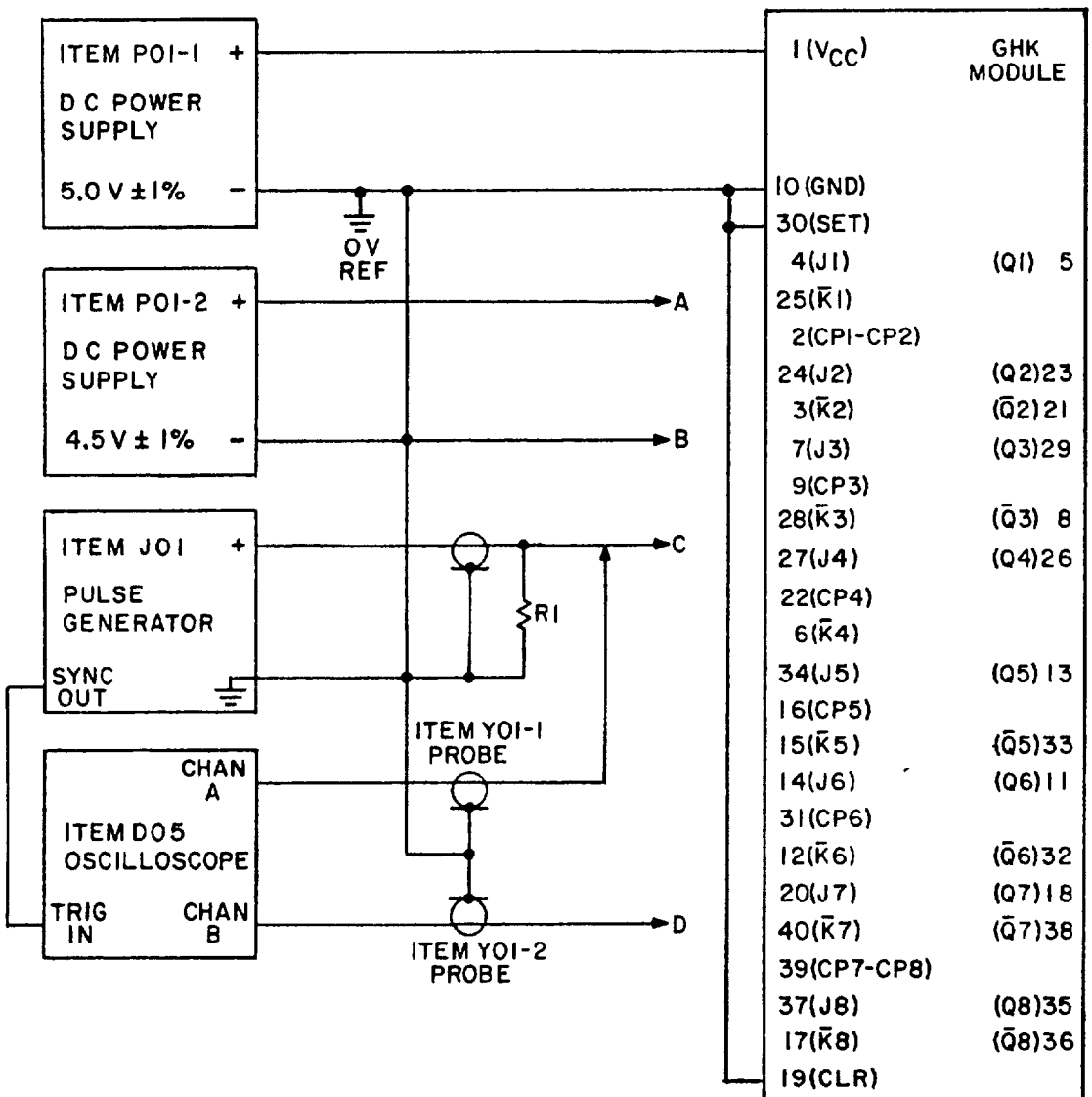
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R5	Q5	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	34	15	16	13	4	44	ns	20,21
R6	Q6		14	12	31	11				
R7	Q7		20	40	39	18				
R8	Q8		37	17	39	35				
R9	$\overline{Q2}$		24	3	2	21				
R10	$\overline{Q3}$		7	28	9	8				
R11	$\overline{Q5}$		34	15	16	33				
R12	$\overline{Q6}$		14	12	31	32				
R13	$\overline{Q7}$		20	40	39	38				
R14	$\overline{Q8}$	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	37	17	39	36	4	44		
R15	Q1	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	4	25	2	5	5	54		
R16	Q2		24	3	2	23				
R17	Q3		7	28	9	29				
R18	Q4		27	6	22	26				
R19	Q5		34	15	16	13				
R20	Q6	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	14	12	31	11	5	54	ns	20,21

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*TABLE XV. Propagation delay time from clock test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R21	Q7	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	20	40	39	18	5	54	ns	20,21
R22	Q8	↓	37	17	39	35	↓	↓	↓	↓
R23	$\overline{Q2}$		24	3	2	21				
R24	$\overline{Q3}$		7	28	9	8				
R25	$\overline{Q5}$		34	15	16	33				
R26	$\overline{Q6}$		14	12	31	32				
R27	$\overline{Q7}$		20	40	39	38				
R28	$\overline{Q8}$	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	37	17	39	36	5	54	ns	20,21

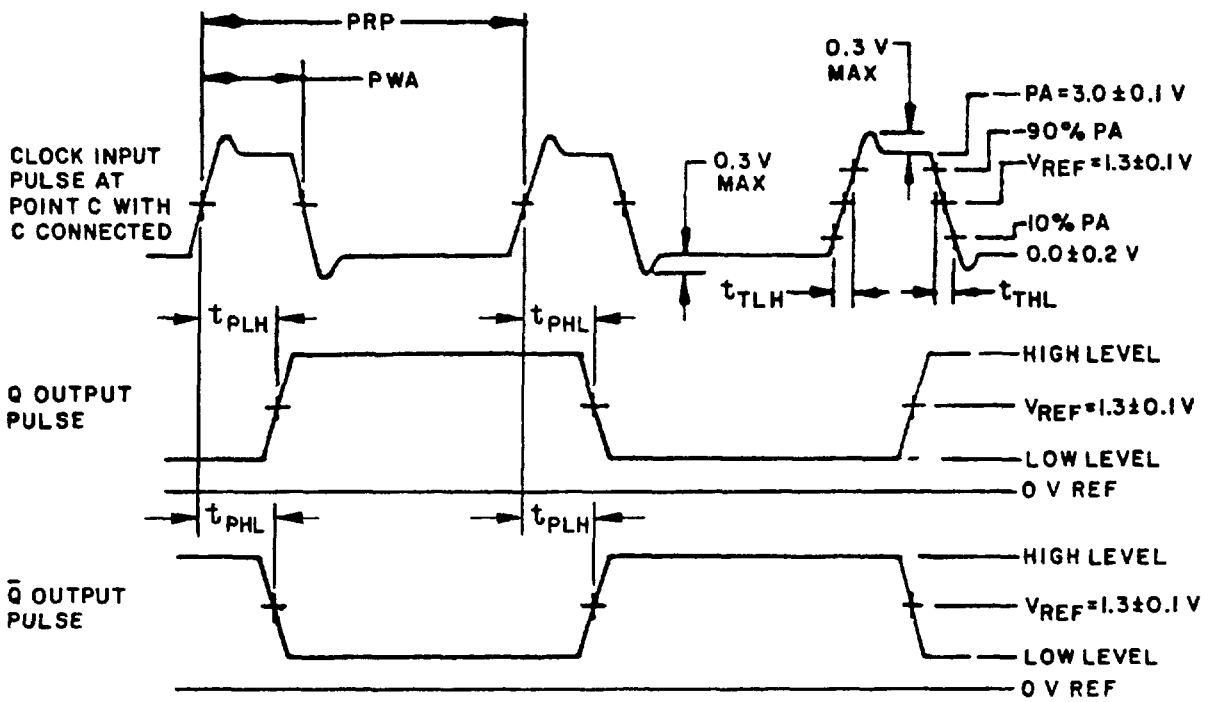
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load outputs per figure 14.

FIGURE 20. Propagation delay time from clock test circuit for table XV.

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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 250 ±50 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements.
3. All voltages are referenced to ground.

FIGURE 21. Propagation delay time from clock test circuit waveforms for table XV.

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* TABLE XVI. Propagation delay time from set and clear test and minimum set and clear pulse width check.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	Q1	Propagation delay time from set (t_{PLH}), C_L = 50 pF	30	19	5	5	55	ns	22,23
S2	Q2				23				
S3	Q3				29				
S4	Q4				26				
S5	Q5				13				
S6	Q6				11				
S7	Q7				18				
S8	Q8	Propagation delay time from set (t_{PLH}), C_L = 50 pF			35		55		
S9	Q1	Propagation delay time from set (t_{PLH}), C_L = 150 pF			5		65		
S10	Q2				23				
S11	Q3				29				
S12	Q4				26				
S13	Q5				13				
S14	Q6				11				
S15	Q7				18				
S16	Q8	Propagation delay time from set (t_{PLH}), C_L = 150 pF	30	19	35	5	65	ns	22,23

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* TABLE XVI. Propagation delay time from set and clear test and minimum set and clear pulse width check - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	$\overline{Q}2$	Propagation delay time from set (t_{PHL}), C_L = 50 pF	30	19	21	6	65	ns	22,23
T2	$\overline{Q}3$				8				
T3	$\overline{Q}5$				33				
T4	$\overline{Q}6$				32				
T5	$\overline{Q}7$				38				
T6	$\overline{Q}8$	Propagation delay time from set (t_{PHL}), C_L = 50 pF			36	6	65		
T7	$\overline{Q}2$	Propagation delay time from set (t_{PHL}), C_L = 150 pF			21	7	75		
T8	$\overline{Q}3$				8				
T9	$\overline{Q}5$				33				
T10	$\overline{Q}6$				32				
T11	$\overline{Q}7$				38				
T12	$\overline{Q}8$	Propagation delay time from set (t_{PHL}), C_L = 150 pF	30	19	36	7	75	ns	22,23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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* TABLE XVI. Propagation delay time from set and clear test and minimum set and clear pulse width check - Continued.

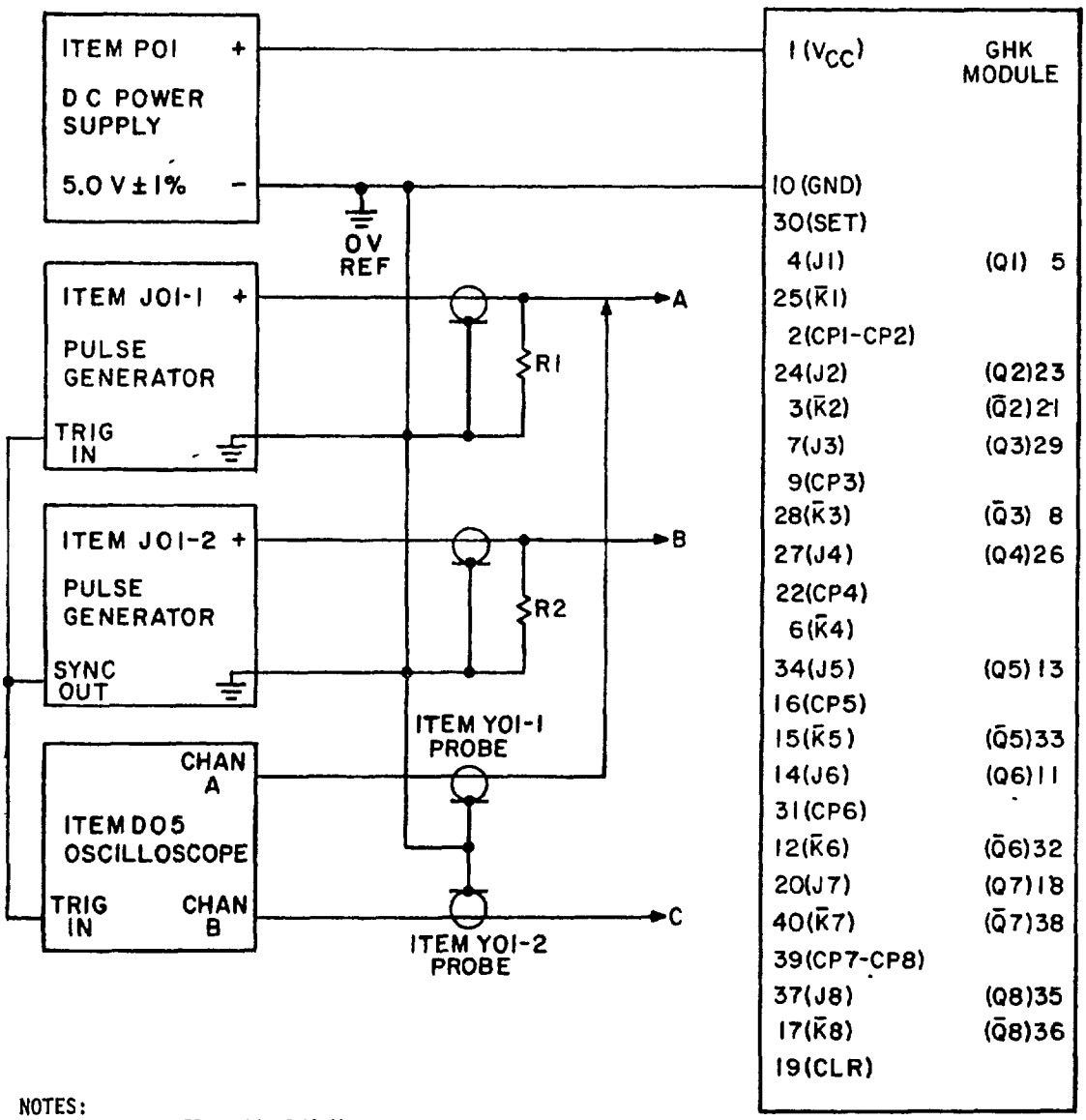
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U1	$\overline{Q2}$	Propagation delay time from clear (t_{PLH}), C_L = 50 pF	19	30	21	5	50	ns	22,24
U2	$\overline{Q3}$				8				
U3	$\overline{Q5}$				33				
U4	$\overline{Q6}$				32				
U5	$\overline{Q7}$				38				
U6	$\overline{Q8}$	Propagation delay time from clear (t_{PLH}), C_L = 50 pF			36	5	50		
U7	$\overline{Q2}$	Propagation delay time from clear (t_{PLH}), C_L = 150 pF			21	6	60		
U8	$\overline{Q3}$				8				
U9	$\overline{Q5}$				33				
U10	$\overline{Q6}$				32				
U11	$\overline{Q7}$				38				
U12	$\overline{Q8}$	Propagation delay time from clear (t_{PLH}), C_L = 150 pF			36		60		
V1	Q1	Propagation delay time from clear (t_{PHL}), C_L = 50 pF			5		63		
V2	Q2	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	19	30	23	6	63	ns	22,24

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*TABLE XVI. Propagation delay time from set and clear test and minimum set and clear pulse width check - Continued.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
V3	Q3	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	19	30	29	6	63	ns	22,24
V4	Q4				26				
V5	Q5				13				
V6	Q6				11				
V7	Q7				18				
V8	Q8	Propagation delay time from clear (t_{PHL}), C_L = 50 pF			35	6	63		
V9	Q1	Propagation delay time from clear (t_{PHL}), C_L = 150 pF			5	7	73		
V10	Q2				23				
V11	Q3				29				
V12	Q4				26				
V13	Q5				13				
V14	Q6				11				
V15	Q7				18				
V16	Q8	Propagation delay time from clear (t_{PHL}), C_L = 150 pF	19	30	35	7	73	ns	22,24

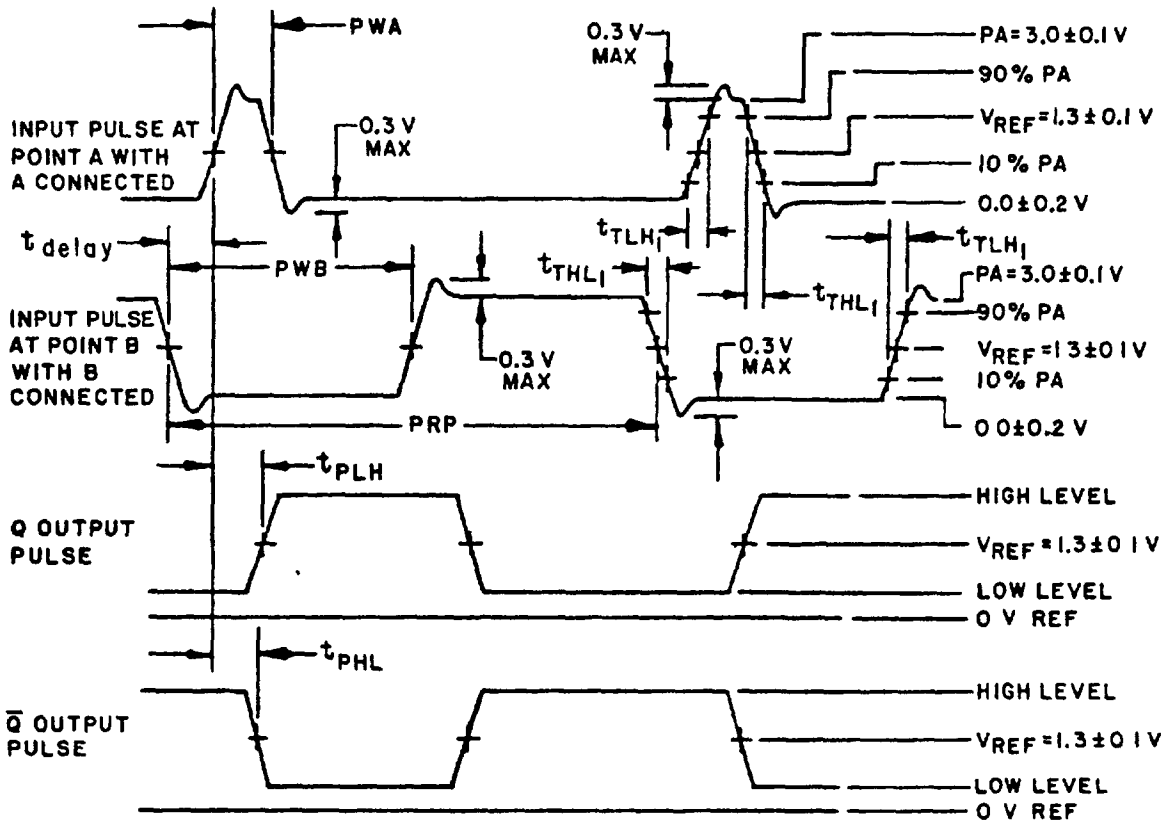
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 14.

FIGURE 22. Propagation delay time from set and clear test and minimum set and clear pulse width check test circuit for table XVI.

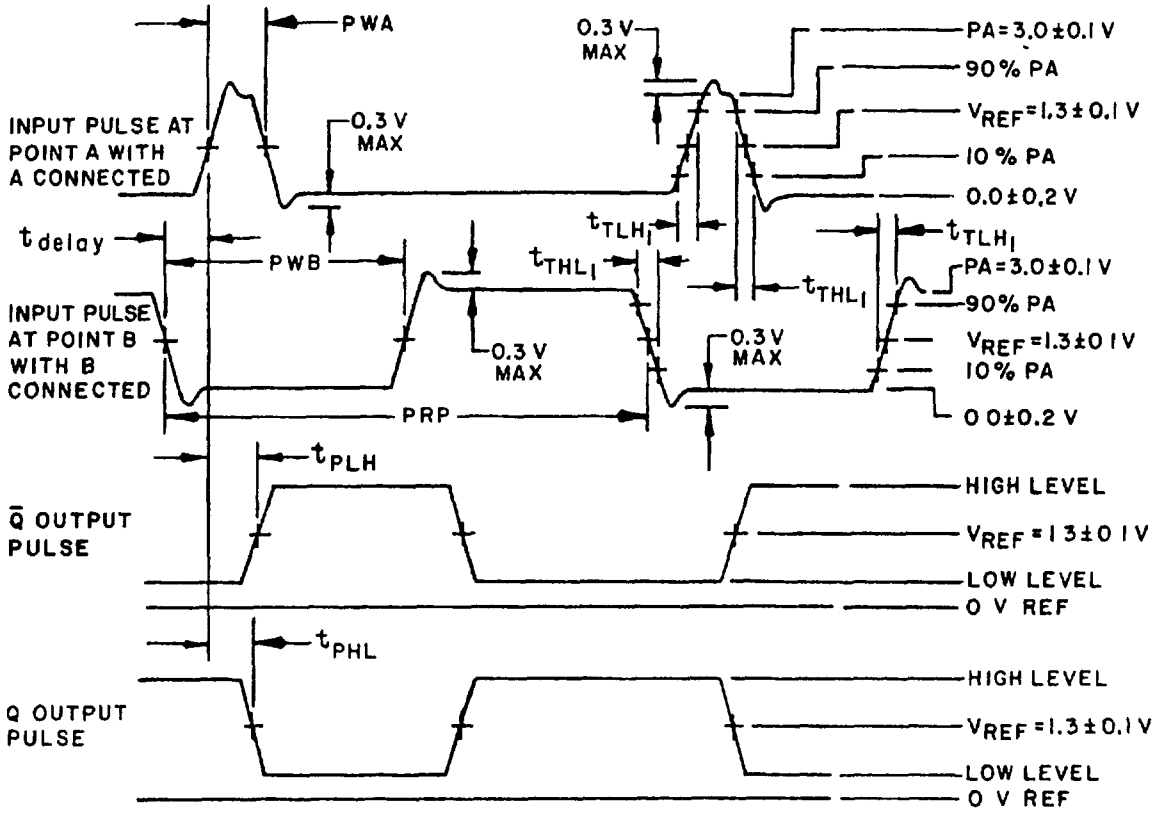
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NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 1,000 ±100 ns.
   PWB = 250 ±25 ns.
   $t_{delay} = 50 \pm 5$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. Adjust item J01-1 to obtain PWA = $25 \, ^{+1.0}_{-0}$ ns (minimum set and clear pulse width).
   b. The pulse width requirements must be verified before making measurements.
4. Verify that all outputs have a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
5. All voltages are referenced to ground.

FIGURE 23. Propagation delay time from set test and minimum set pulse width check test circuit waveforms for table XVI.

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NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3.
   $t_{TLHl} = t_{THLl} = 6 \pm 1$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 1,000 ±100 ns.
   PWB = 250 ±25 ns.
   $t_{delay} = 50 \pm 5$ ns.
   $t_{TLHl} = t_{THLl} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. Adjust item J01-1 to obtain PWA = $25\ ^{+1.0}_{-0}$ ns (minimum set and clear pulse width).
   b. The pulse width requirements must be verified before making measurements.
4. Verify that all outputs have a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
5. All voltages are referenced to ground.

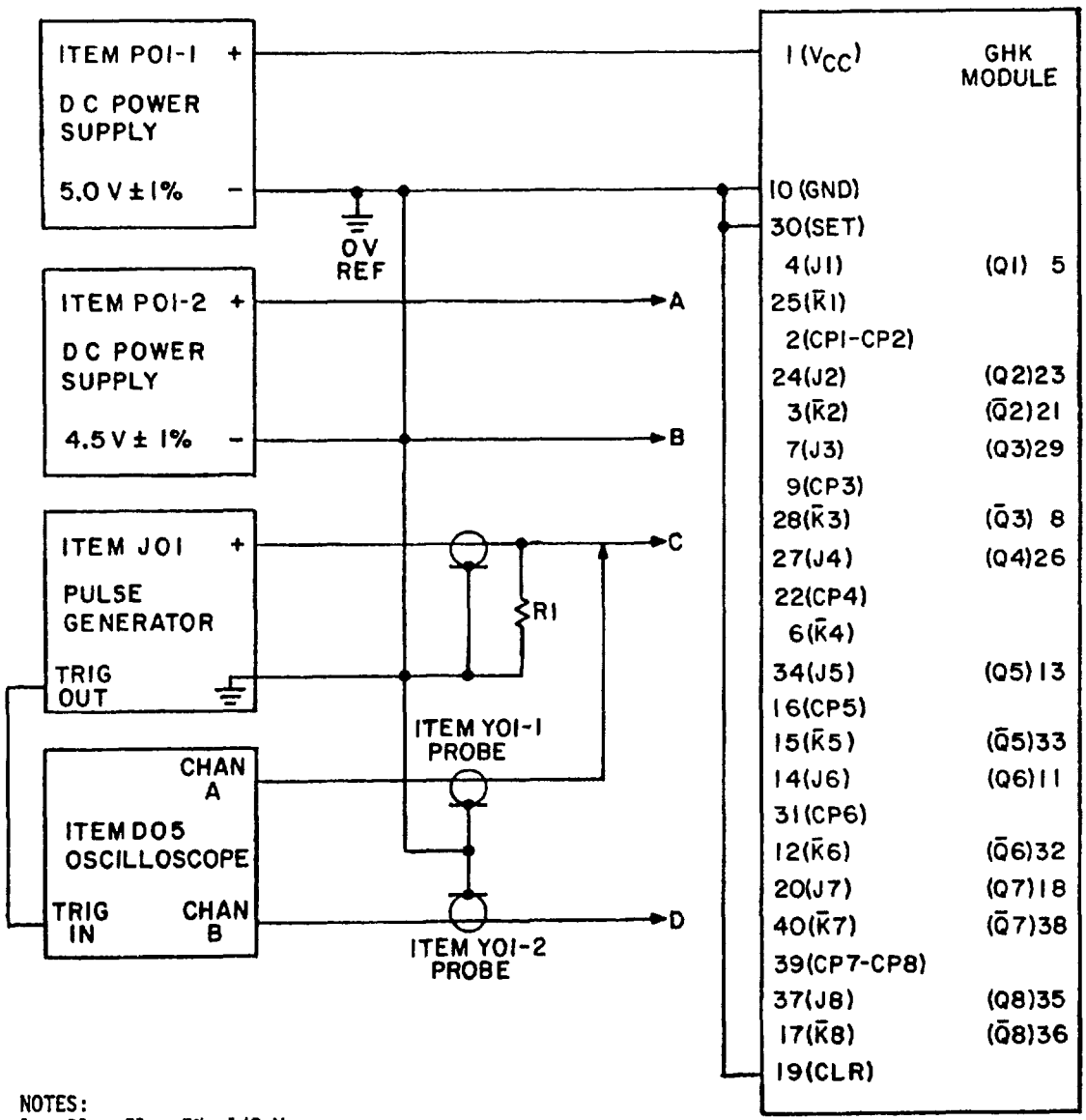
FIGURE 24. Propagation delay time from clear test and minimum clear pulse width check test circuit waveforms for table XVI.

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TABLE XVII. Maximum operating frequency check and minimum clock pulse width check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	Q1 Maximum operating frequency check (measuring output pulse period - t_{PW}), C_L = 50 pF	4	25	2	5	75	85	ns	25,26
W2	Q2	24	3	2	23				
W3	Q3	7	28	9	29				
W4	Q4	27	6	22	26				
W5	Q5	34	15	16	13				
W6	Q6	14	12	31	11				
W7	Q7	20	40	39	18				
W8	Q8	37	17	39	35				
W9	$\overline{Q2}$	24	3	2	21				
W10	$\overline{Q3}$	7	28	9	8				
W11	$\overline{Q5}$	34	15	16	33				
W12	$\overline{Q6}$	14	12	31	32				
W13	$\overline{Q7}$	20	40	39	38				
W14	$\overline{Q8}$ Maximum operating frequency check (measuring output pulse period - t_{PW}), C_L = 50 pF	37	17	39	36	75	85	ns	25,26

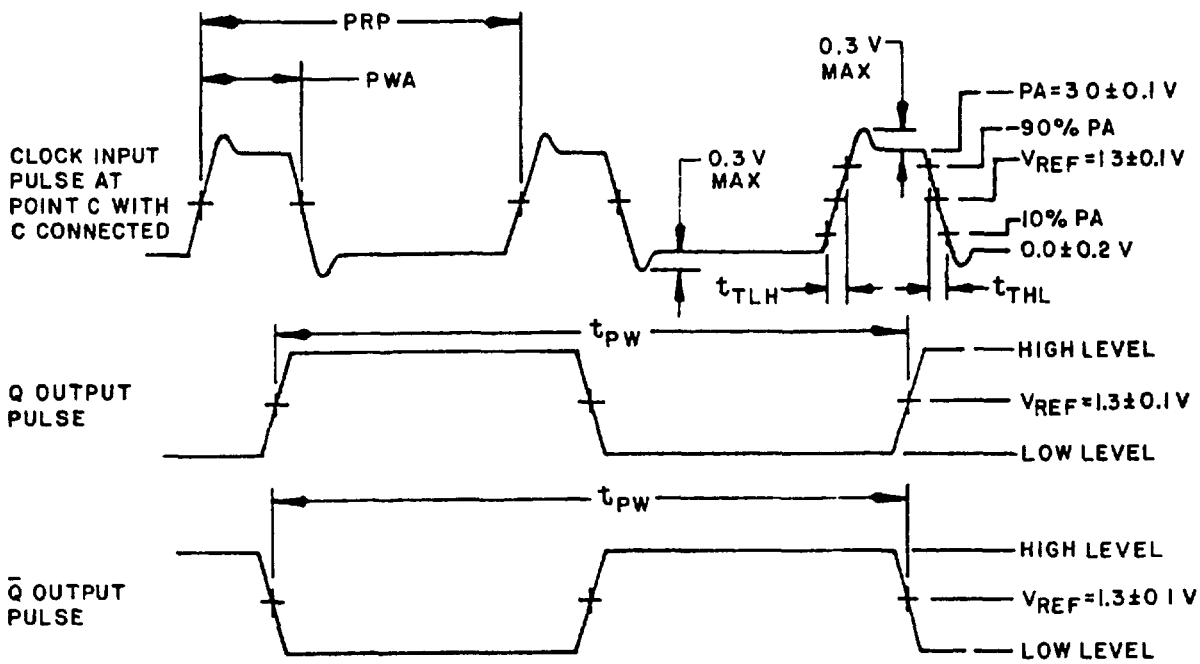
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 14.

FIGURE 25. Maximum operating frequency check and minimum clock pulse width check test circuit for table XVII.

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NOTES:
1. Clock input pulse parameters:

   PRP = $40^{+2.0}_{-0}$ ns ($25.0^{+0}_{-1.2}$ MHz).

   PWA = $25^{+1.0}_{-0}$ ns.

   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
2. Verify that all outputs have a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
3. All voltages are referenced to ground.

*FIGURE 26. Maximum operating frequency and minimum clock pulse width check test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	Q1 Output crosstalk, low level	7,14,20, 24,27,34, 37	3,4,6,12, 15,17,25, 28,40	5	0	400	mV	27,28
X2	Q2	4,7,14,20, 27,34,37	3,6,12, 15,17,24, 25,28,40	23				
X3	Q3	4,14,20, 24,27,34, 37	3,6,7,12, 15,17,25, 28,40	29				
X4	Q4	4,7,14,20, 24,34,37	3,6,12, 15,17,25, 27,28,40	26				
X5	Q5	4,7,14,20, 24,27,37	3,6,12, 15,17,25, 28,34,40	13				
X6	Q6	4,7,20,24, 27,34,37	3,6,12, 14,15,17, 25,28,40	11				
X7	Q7	4,7,14,24, 27,34,37	3,6,12, 15,17,20, 25,28,40	18				
X8	Q8	4,7,14,20, 24,27,34	3,6,12, 15,17,25, 28,37,40	35				
X9	$\overline{Q2}$ Output crosstalk, low level	3,4,7,14, 20,24,27, 34,37	6,12,15, 17,25,28, 40	21	0	400	mV	27,28

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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X10	$\overline{Q}$3 Output crosstalk, low level	4,7,14,20, 24,27,28, 34,37	3,6,12, 15,17,25, 40	8	0	400	mV	27,28
X11	$\overline{Q}$5	4,7,14,15, 20,24,27, 34,37	3,6,12, 17,25,28, 40	33				
X12	$\overline{Q}$6	4,7,12,14, 20,24,27, 34,37	3,6,15 17,25,28, 40	32				
X13	$\overline{Q}$7	4,7,14,20, 24,27,34, 37,40	3,6,12, 15,17,25, 28	38				
X14	$\overline{Q}$8 Output crosstalk, low level	4,7,14,17, 20,24,27, 34,37	3,6,12, 15,25,28, 40	36				
Y1	Q1 Output crosstalk, high level	4,7,14,20, 24,25,27, 34,37	3,6,12, 15,17,28, 40	5				
Y2	Q2	3,4,7,14, 20,24,27, 34,37	6,12,15, 17,25,28 40	23				
Y3	Q3	4,7,14,20, 24,27,28, 34,37	3,6,12, 15,17,25, 40	29				
Y4	Q4 Output crosstalk, high level	4,6,7,14, 20,24,27, 34,37	3,12,15, 17,25,28, 40	26	0	400	mV	27,28

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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

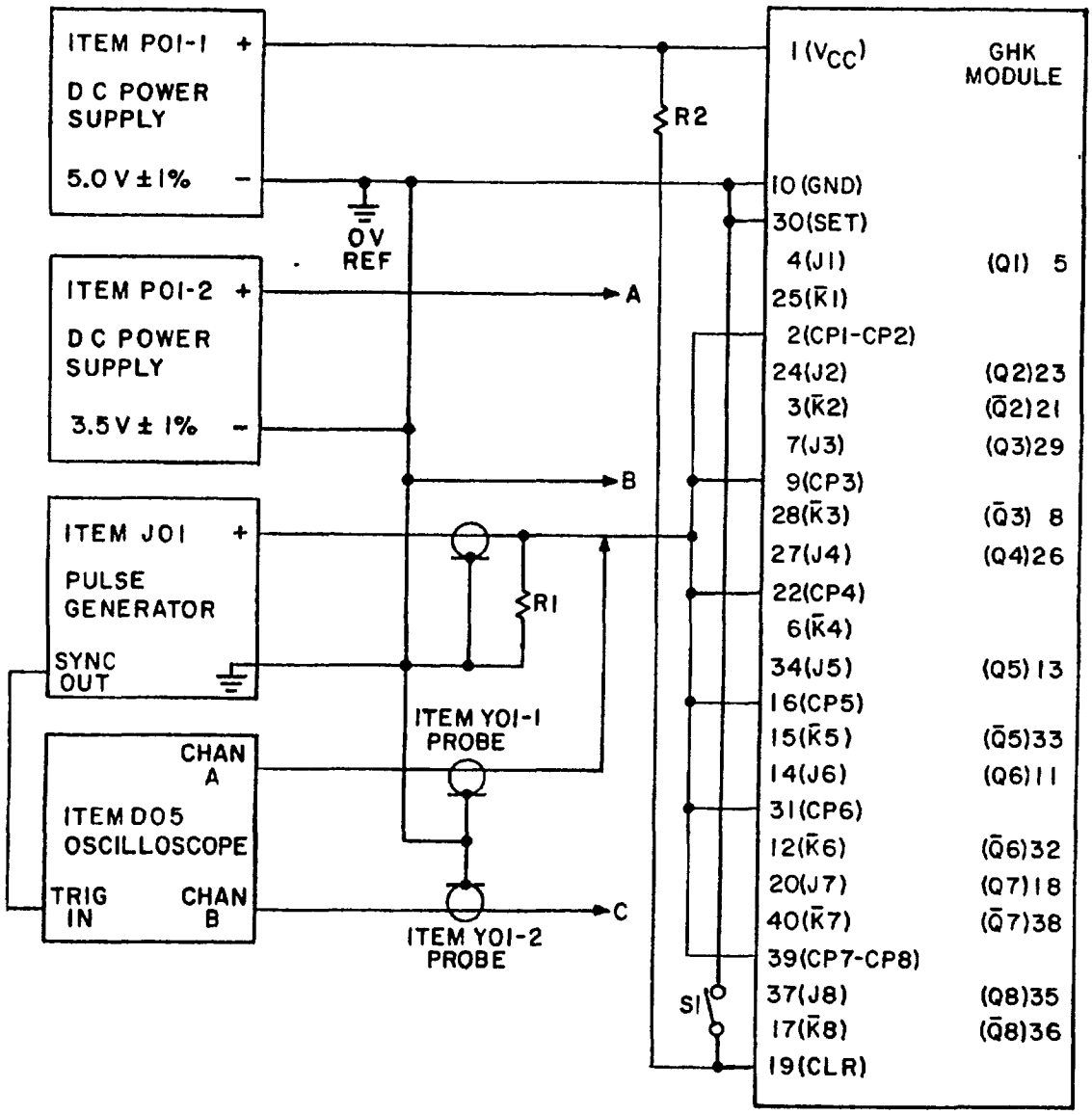
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y5	Q5 Output crosstalk, high level	4,7,14,15, 20,24,27, 34,37	3,6,12, 17,25,28, 40	13	0	400	mV	27,28
Y6	Q6	4,7,12,14, 20,24,27, 34,37	3,6,15, 17,25,28, 40	11				
Y7	Q7	4,7,14,20, 24,27,34, 37,40	3,6,12, 15,17,25, 28	18				
Y8	Q8	4,7,14,17, 20,24,27, 34,37	3,6,12, 15,25,28, 40	35				
Y9	$\overline{Q2}$	4,7,14,20, 27,34,37	3,6,12, 15,17,24, 25,28,40	21				
Y10	$\overline{Q3}$	4,14,20, 24,27,34, 37	3,6,7,12, 15,17,25, 28,40	8				
Y11	$\overline{Q5}$	4,7,14,20, 24,27,37	3,6,12, 15,17,25, 28,34,40	33				
Y12	$\overline{Q6}$	4,7,20,24, 27,34,37	3,6,12, 14,15,17, 25,28,40	32				
Y13	$\overline{Q7}$ Output crosstalk, high level	4,7,14,24, 27,34,37	3,6,12, 15,17,20, 25,28,40	38	0	400	mV	27,28

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y14	$\overline{Q}8$ Output crosstalk, high level	4,7,14,20, 24,27,34	3,6,12, 15,17,25, 28,37,40	36	0	400	mV	27,28

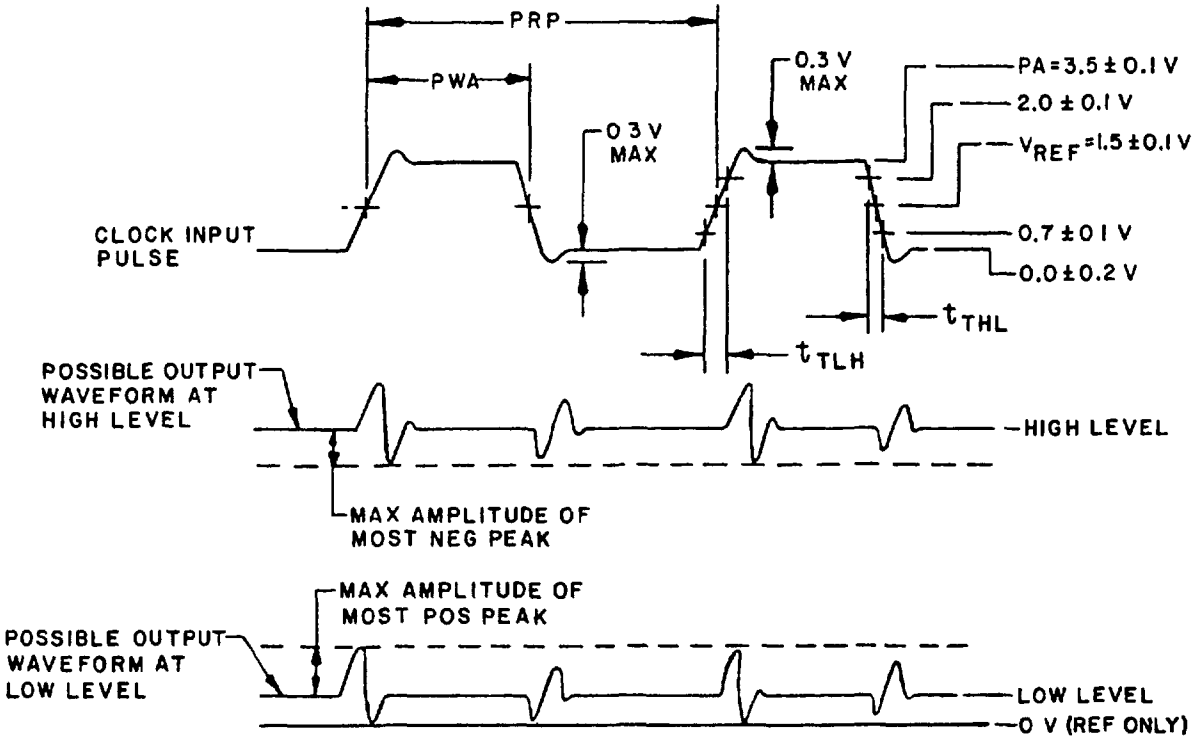
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. Load all outputs with load as specified on figure 14 with the exception that capacitor C_L = 50 pF.
5. After making all circuit connections for a test condition open switch S1 for a minimum of 100 ns then close switch S1.

FIGURE 27. Output crosstalk (low-level and high-level) test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:28Z
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NOTES:
1. Clock input pulse parameters:

   a. PRP = 1,000 ±200 ns
   b. PWA = 500 ±100 ns.
   c. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.7 V and 2.0 V points of the input pulse).
   d. For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making each measurement.

2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 2 inches (50.8 mm).

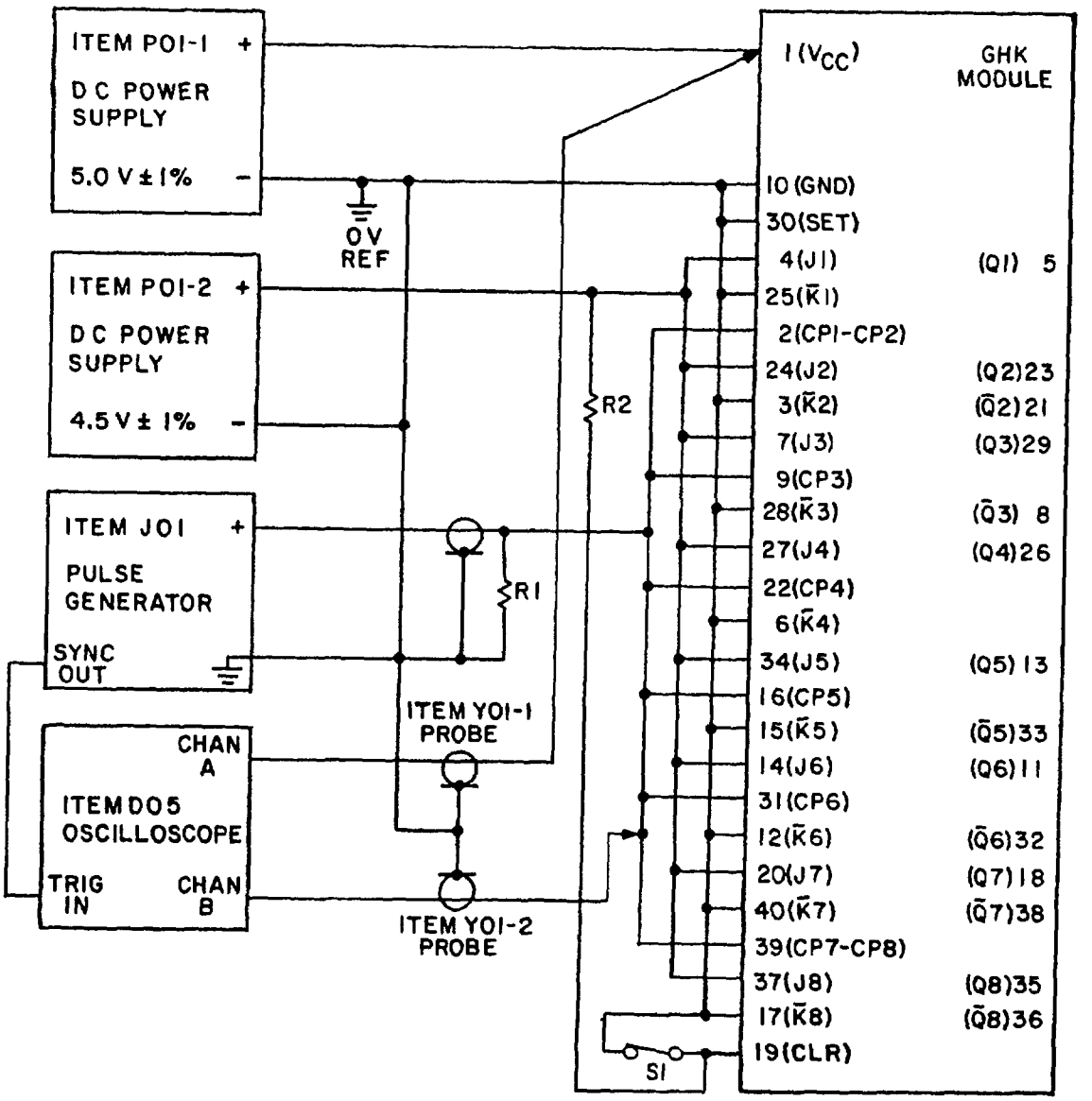
FIGURE 28. Output crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	400	mV	29,30

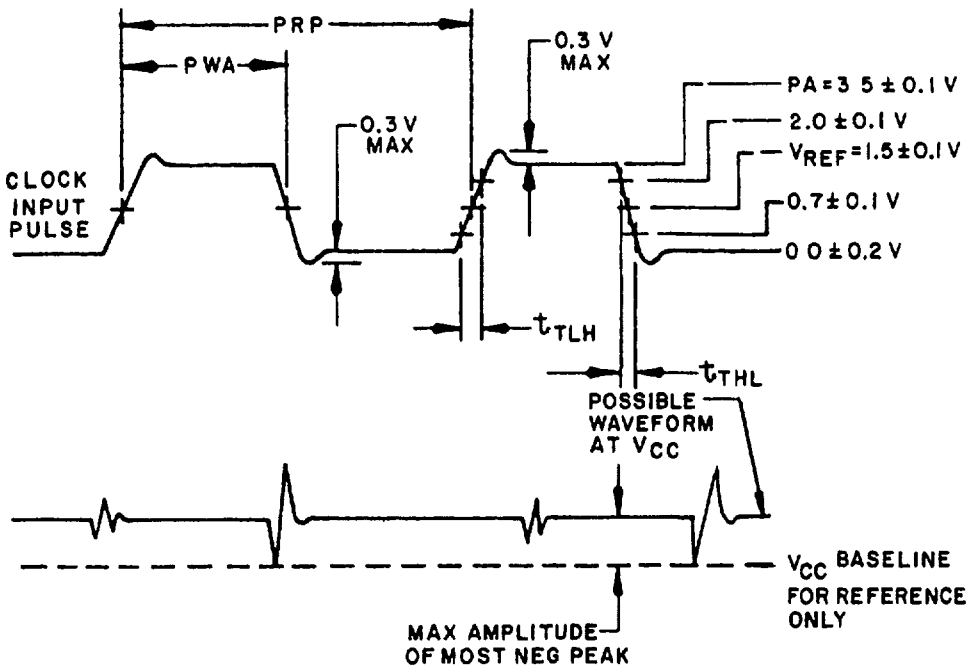
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. After making all circuit connections for the test condition, open switch S1 for a minimum of 100 ns then close switch S1 and make measurement.

FIGURE 29. Power supply transient amplitude test circuit for table XIX.

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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 500 ±100 ns.
   t_{TLH} = t_{THL} = Vary t_{TLH} from 5 to 25 ns and t_{THL} from 3 to 10 ns and record the maximum value of transient amplitude.
2. For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making each measurement.

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	1.6	3.0	µF	31

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* TABLE XXI. Input capacitance test.

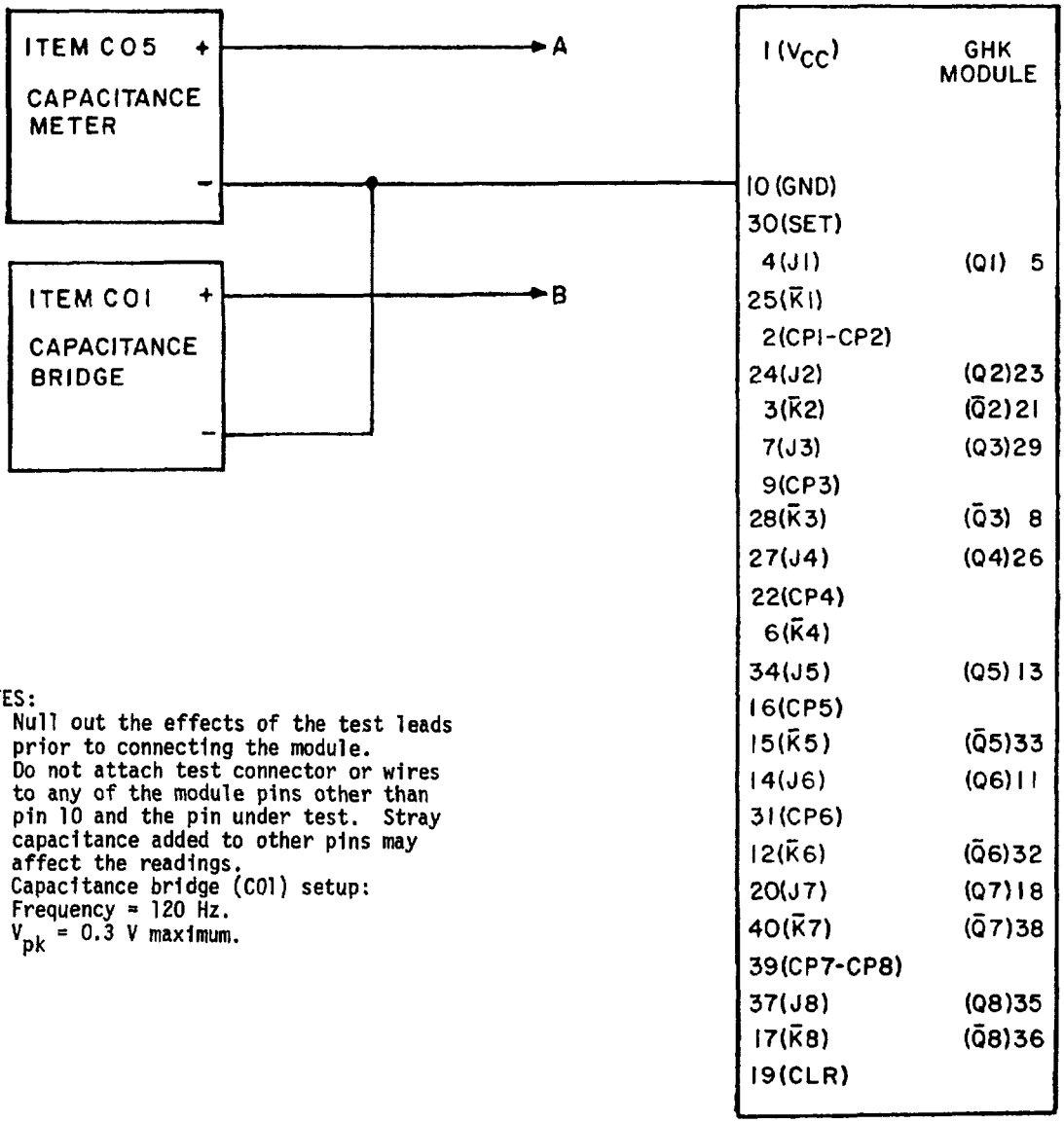
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	CLR Input capacitance	19	Open	5	45	pF	31
BB2	SET	30			30		
BB3	CP1-CP2	2			30		
BB4	CP7-CP8	39			30		
BB5	CP3	9			15		
BB6	CP4	22					
BB7	CP5	16					
BB8	CP6	31					
BB9	J1	4					
BB10	J2	24					
BB11	J3	7					
BB12	J4	27					
BB13	J5	34					
BB14	J6	14					
BB15	J7	20					
BB16	J8	37					
BB17	$\overline{K1}$	25					
BB18	$\overline{K2}$ Input capacitance	3	Open	5	15	pF	31

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* TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB19	$\overline{K3}$ Input capacitance	28	Open	5	15	pF	31
BB20	$\overline{K4}$	6	↓	↓	↓	↓	↓
BB21	$\overline{K5}$	15					
BB22	$\overline{K6}$	12					
BB23	$\overline{K7}$	40					
BB24	$\overline{K8}$ Input capacitance	17	Open	5	15	pF	31

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup: Frequency = 120 Hz. V_{pk} = 0.3 V maximum.

FIGURE 31. Power supply filter and input capacitance test circuit for tables XX and XXI.

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* TABLE XXII. Setup time test.

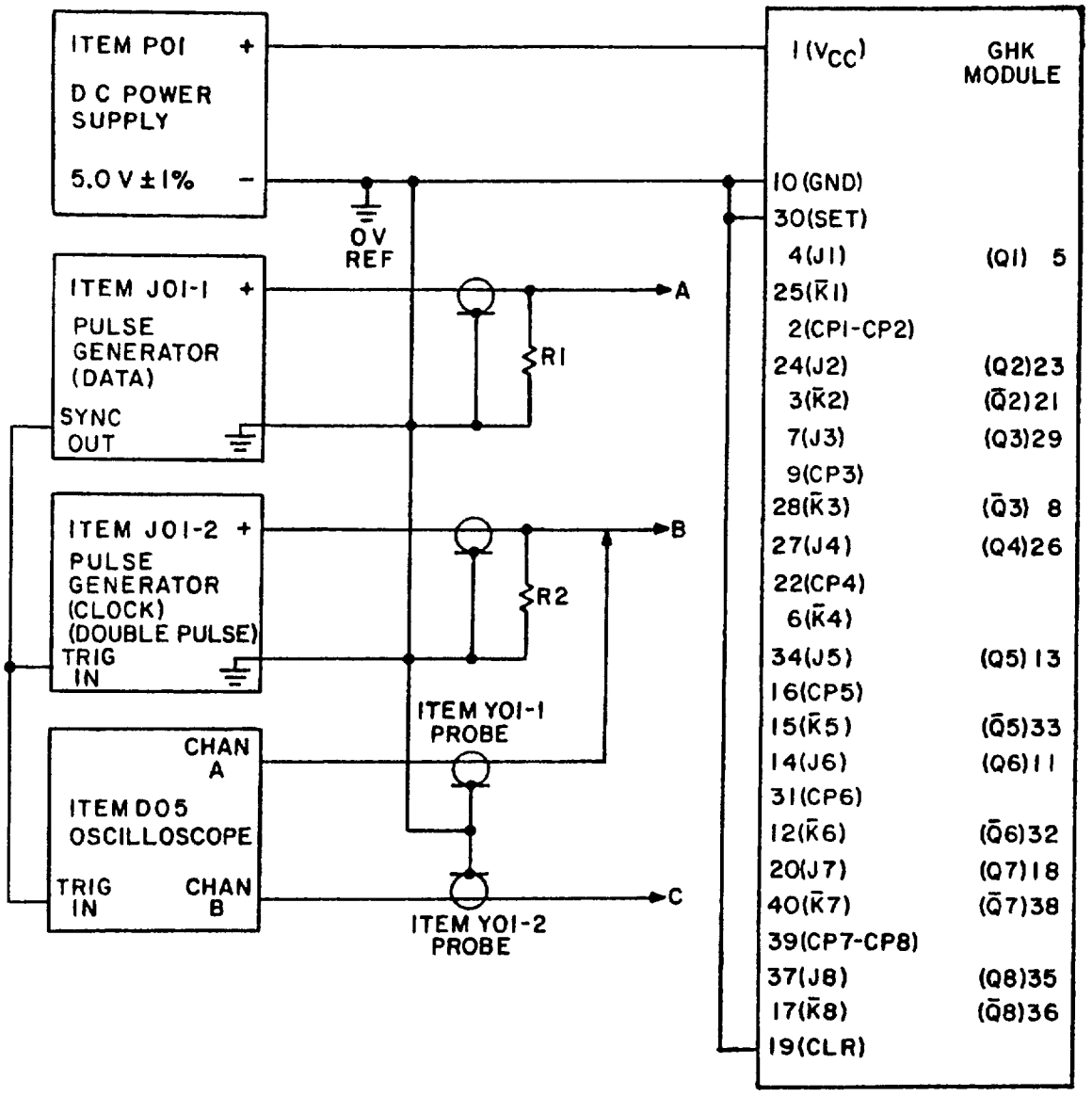
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
CC1	Q1 Setup time, C_L = 50 pF	4,25	2	5	170	230	ns	32,33
CC2	Q2	3,24	2	23				
CC3	Q3	7,28	9	29				
CC4	Q4	6,27	22	26				
CC5	Q5	15,34	16	13				
CC6	Q6	12,14	31	11				
CC7	Q7	20,40	39	18				
CC8	Q8	17,37	39	35				
CC9	$\overline{Q2}$	3,24	2	21				
CC10	$\overline{Q3}$	7,28	9	8				
CC11	$\overline{Q5}$	15,34	16	33				
CC12	$\overline{Q6}$	12,14	31	32				
CC13	$\overline{Q7}$	20,40	39	38				
CC14	$\overline{Q8}$	17,37	39	36				
CC15	Q1	4,25	2	5				
CC16	Q2 Setup time, C_L = 50 pF	3,24	2	23	170	230	ns	32,33

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* TABLE XXII. Setup time test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
CC17	Q3 Setup time, C_L = 50 pF	7,28	9	29	170	230	ns	32,33
CC18	Q4	6,27	22	26				
CC19	Q5	15,34	16	13				
CC20	Q6	12,14	31	11				
CC21	Q7	20,40	39	18				
CC22	Q8	17,37	39	35				
CC23	$\overline{Q2}$	3,24	2	21				
CC24	$\overline{Q3}$	7,28	9	8				
CC25	$\overline{Q5}$	15,34	16	33				
CC26	$\overline{Q6}$	12,14	31	32				
CC27	$\overline{Q7}$	20,40	39	38				
CC28	$\overline{Q8}$ Setup time, C_L = 50 pF	17,37	39	36	170	230	ns	32,33

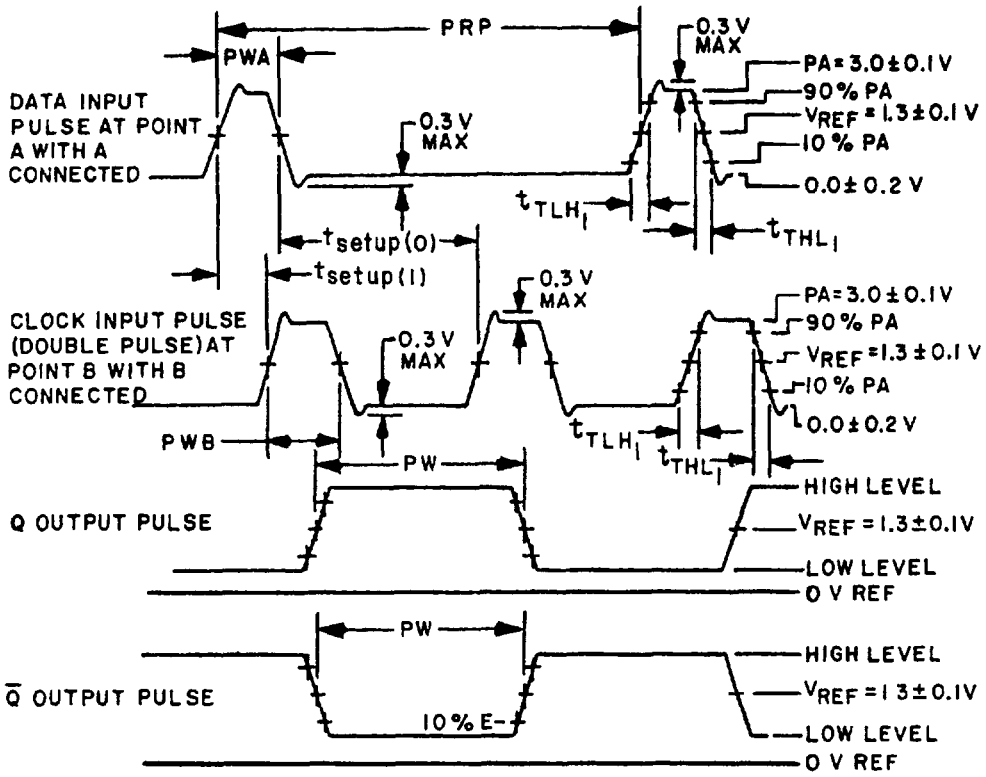
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 14.

FIGURE 32. Setup time test circuit for table XXII.

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NOTES:
1. Data input pulse parameters:

   a. PRP = 400 ±40 ns.
   b. $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
   c. PWA = 200 ±20 ns.

2. Clock input pulse parameters.

   a. Item J01-2 is operated in double pulse mode.
   b. PWB = 100 ±10 ns.
   c. $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.

3. Minimum setup time clock:

   a. For test conditions CC1-CC14: $t_{setup(1)} = 20\ ^{+2}_{-0}$ ns, $t_{setup(0)} = 40 \pm 4$ ns.
   b. For test conditions CC15-CC28: $t_{setup(1)} = 40 \pm 4$ ns, $t_{setup(0)} = 20\ ^{+2}_{-0}$ ns.
   c. The setup time requirements must be verified before making measurements per table XXII.

4. Verify that all outputs have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements per table XXII.
5. All voltages are referenced to ground.

FIGURE 33. Setup time test circuit waveforms for table XXII.

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* TABLE XXIII. Hold time test.

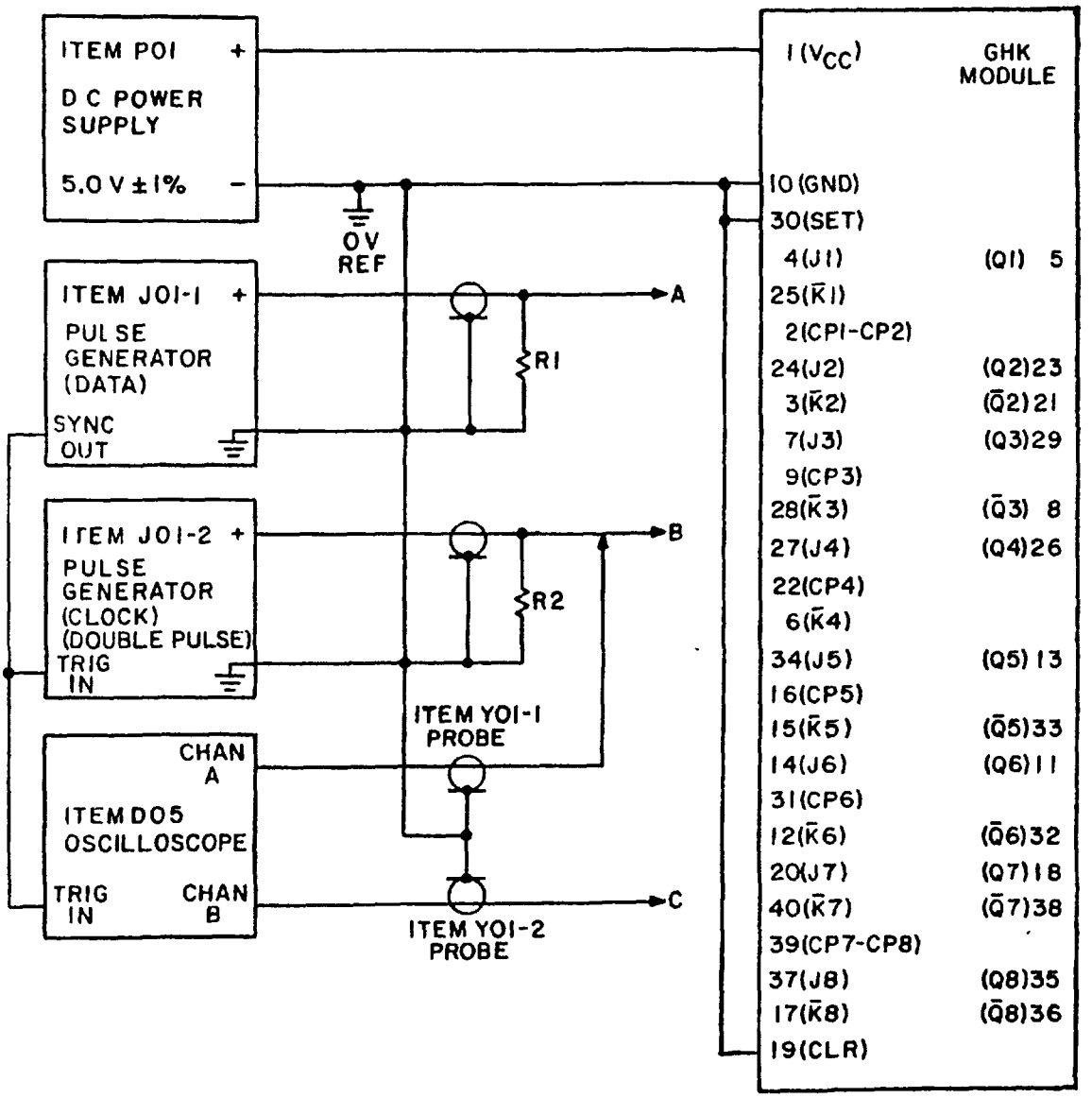
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
DD1	Q1 Hold time C_L = 50 pF	4,25	2	5	170	230	ns	34,35
DD2	Q2	3,24	2	23				
DD3	Q3	7,28	9	29				
DD4	Q4	6,27	22	26				
DD5	Q5	15,34	16	13				
DD6	Q6	12,14	31	11				
DD7	Q7	20,40	39	18				
DD8	Q8	17,37	39	35				
DD9	$\overline{Q2}$	3,24	2	21				
DD10	$\overline{Q3}$	7,28	9	8				
DD11	$\overline{Q5}$	15,34	16	33				
DD12	$\overline{Q6}$	12,14	31	32				
DD13	$\overline{Q7}$	20,40	39	38				
DD14	$\overline{Q8}$	17,37	39	36				
DD15	Q1	4,25	2	5				
DD16	Q2 Hold time, C_L = 50 pF	3,24	2	23	170	230	ns	34,35

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* TABLE XXIII. Hold time test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
DD17	Q3 Hold time C_L = 50 pF	7,28	9	29	170	230	ns	34,35
DD18	Q4	6,27	22	26				
DD19	Q5	15,34	16	13				
DD20	Q6	12,14	31	11				
DD21	Q7	20,40	39	18				
DD22	Q8	17,37	39	35				
DD23	$\overline{Q}$2	3,24	2	21				
DD24	$\overline{Q}$3	7,28	9	8				
DD25	$\overline{Q}$5	15,34	16	33				
DD26	$\overline{Q}$6	12,14	31	32				
DD27	$\overline{Q}$7	20,40	39	38				
DD28	$\overline{Q}$8 Hold time C_L = 50 pF	17,37	39	36	170	230	ns	34,35

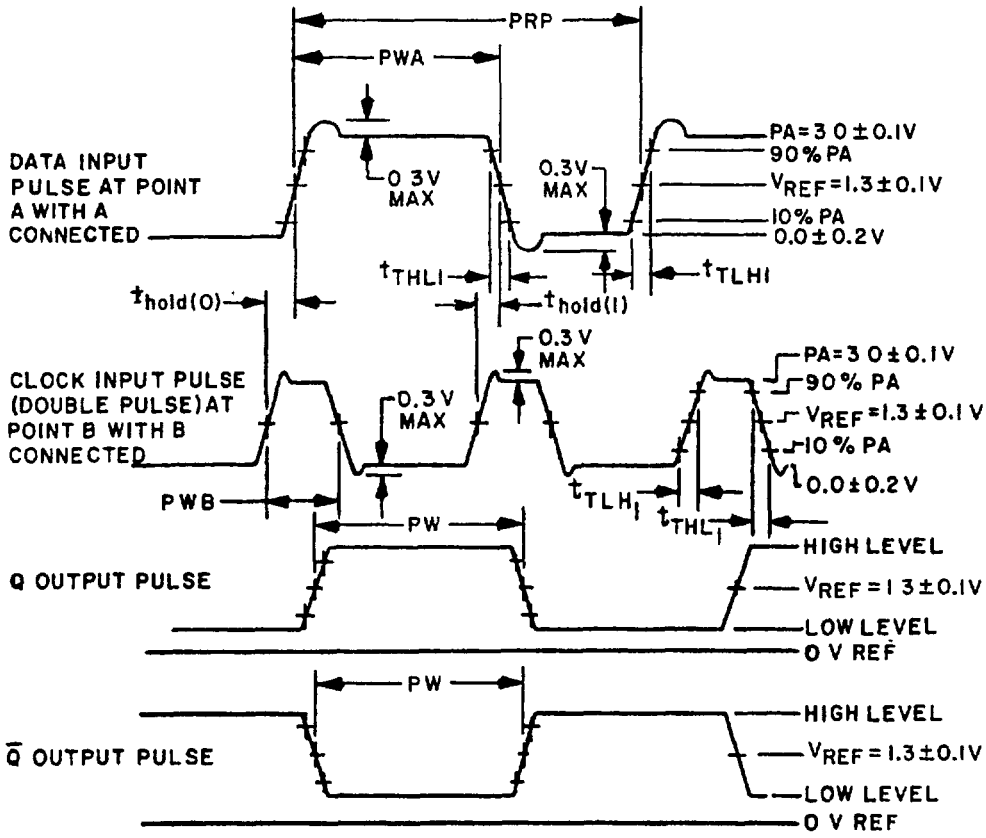
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 14.

FIGURE 34. Hold time test circuit for table XXIII.

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NOTES:
1. Data input pulse parameters:
   a. PRP = 400 ±40 ns.
   b. $t_{TLH1} = t_{THL1}$ = 6 ±1 ns.
   c. PWA = 200 ±20 ns.
2. Clock input pulse parameters:
   a. Item J01-2 is operated in double pulse mode.
   b. PWB = 100 ±10 ns.
   c. $t_{TLH1} = t_{THL1}$ = 6 ±1 ns.
3. Minimum hold time check:
   a. For test conditions DD1-DD14: $t_{hold(1)} = 5\ 0\ ^{+2}_{-0}$ ns, $t_{hold(0)}$ = 20 ±2 ns.
   b. For test conditions DD15-DD28: $t_{hold(1)}$ = 20 ±2 ns, $t_{hold(0)} = ^{+2}_{-0}$ ns.
   c. The hold time requirements must be verified before making measurements per table XXIII.
4. Verify that all outputs have a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements per table XXIII.
5. All voltages are referenced to ground.

FIGURE 35. Hold time test circuit waveforms for table XXIII.

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